



Multi-objective sustainable closed-loop agricultural supply chain optimization under uncertainty respecting water circularity: olive products

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Abstract

Driven by economic fluctuation and environmental concerns, the agricultural sector urgently requires integrated closed-loop supply chain (CLSC) models that balance profitability with sustainability. Current literature often overlooks simultaneous economic and environmental objectives, particularly concerning water circularity and reverse logistics in the olive industry. This study addresses this gap by proposing a novel bi-objective mixed integer linear programming model designed to minimize both the total supply chain cost (encompassing procurement, facility setup, operation, storage, and transportation) and total water consumption. Uncertainty regarding product yield, influenced by factors like climate and soil, is systematically incorporated using the possibilistic programming approach. The comprehensive network explicitly models forward flows, reverse logistics, package recovery, and rich wastewater utilization. This complex mathematical problem is solved using four rigorously tuned multi-objective metaheuristics (NSGA-II, NPGA, MOEA/D, and MOEA/IL). Applied to a real-world olive network case study, the model demonstrated major reductions in logistics costs, waste production, and water use. A critical finding reveals that the most impactful leverage point for water and cost efficiency lies upstream in cultivation practices (e.g., irrigation management and pest control), rather than downstream industrial recovery, providing actionable guidance for sustainable supply chain management. This research delivers a unified decision framework for water-aware CLSCs under uncertainty.

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1 Introduction

The agricultural sector encounters challenges with increasing demand-market constraints, the need to lower food waste by integrating reverse logistics, and a growing recognition that linear supply chains would waste valuable by-products and natural resources (Liao et al., 2020). However, the agricultural supply chain literature reveals a critical disconnect. Despite its direct impact on crop yield, operational cost, and environmental sustainability, water consumption, a foundational constraint in agricultural production, remains conspicuously absent from most integrated CLSC models. Moreover, the existing Closed-Loop Supply Chain (CLSC) applications in agriculture predominantly adopt forward-focused optimization, treating reverse flows and waste valorization as secondary concerns (Nagurney & Toyasaki, 2005).

Agriculture spans an extraordinary range of crops and supply networks, from citrus and rice to wheat and fresh fruits, each demanding a tailored logistics strategy. This diversity has inspired extensive research into supply chain design (Waqas et al., 2023). The sector's complexity continues to attract attention because of rising crop values, reported by agencies such as the USDA and US Census Bureau (Fig. 1), signal an urgent need for smarter planning that boosts profitability while curbing costs. Yet, one concern is particularly prominent: water management. Although certain crops tolerate drought, irrigation practices profoundly shape yield and quality (Attar et al., 2025a). Too little water depresses productivity, while too much invites pests and disease. Thus, irrigation is not merely an agronomic concern, it is a strategic variable that ripples through every tier of the supply chain. Designing networks that account for water use is therefore essential, as it influences both environmental sustainability and economic performance. Surprisingly, this dimension has often been overlooked in CLSC models, even though controlling water at the source could unlock significant gains downstream.

Among the many crops shaping global agriculture, olives stand out for their dual role as a staple food and a source of premium oil, making their supply chain uniquely complex. Unlike grains or citrus, olives demand meticulous handling across multiple stages: early harvest for table olives and late harvest for oil extraction, each with distinct water requirements and processing routes. This complexity amplifies the challenge of designing sustainable networks for this subsector: irrigation practices influence not only yield but also fruit quality, which cascades into processing efficiency, packaging needs, and reverse flows for waste recovery.

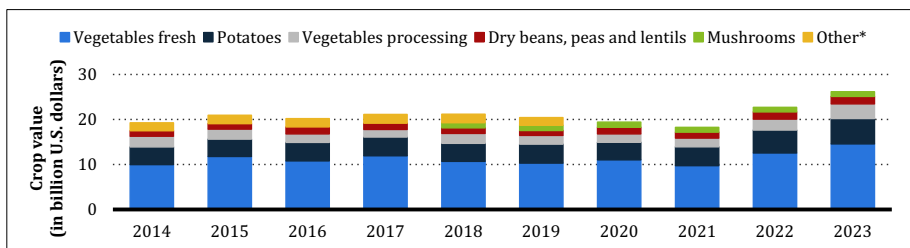


Fig. 1 Crop value of pulses and vegetables in the United States from 2014 to 2023, by type (source: Statista.com)

In regions where water scarcity is acute (such as Mediterranean and Middle Eastern olive belts) the stakes are even higher. Thus, a closed-loop supply chain that integrates forward logistics with reverse flows for packaging reuse and wastewater valorization would offer a promising path to reconcile profitability with environmental stewardship.

Designing such a closed-loop supply chain for olives is not just for academic practice, it is also a practical necessity. Olives are versatile: they serve as table fruit, yield oil for culinary and industrial uses, and generate by-products that can be transformed into fertilizers or animal feed. This multi-purpose nature creates a web of interconnected flows, from gardens to mills, packaging centers, and recovery facilities. Unlike many crops, olives demand a network that accommodates both forward and reverse logistics, ensuring that residuals and packaging re-enter the cycle rather than becoming waste. Nonetheless, despite olives' prominence in Mediterranean economies and their growing global demand, no comprehensive CLSC model has been developed for this industry. Our study addresses this gap for comprehensive models while embedding water fairness into the design—a novel dimension that aligns economic efficiency with environmental responsibility and circularity in this sector.

To achieve this, we propose a multi-objective CLSC model that simultaneously minimizes total logistics costs and water consumption. Here, natural uncertainties compound the challenge: climate, soil conditions, and pest dynamics fluctuate olive yields, rendering deterministic planning unrealistic. We tackle this complexity using possibilistic programming and advanced metaheuristic methods to generate robust solutions. Thus, the main contributions of this study can be summarized as follows: (1) developing a comprehensive model for the olive industry with all major stakeholders, (2) considering the different harvest stages and the corresponding productivity, (3) acknowledging the effect of the climate condition on water consumption, (4) packaging circularity and correlated waste management, (5) water recovery and circularity, (6) stage-based harvest modeling and optimization, (7) multiple, tuned algorithms to tackle mathematical complexity, and finally (8) the implementation of the proposed framework in a real case study for practical managerial insights.

The rest of the paper is organized as follows: Sect. 2 presents a thorough review of related literature and trends in the Agricultural Supply Chain (ASC) and CLSC. The system description, assumptions, and the proposed bi-objective network design problem for the olive industry are covered in Sect. 3. The mathematical model and uncertainty framework are presented in Sect. 4. Section 5 focuses on the deployed solution methods and proposes a suitable encoding plan for the utilized algorithms. The proposed approach is also investigated by considering various performance metrics in Sect. 5. Section 6 provides further sensitivity analysis and managerial implications. Lastly, Sect. 7 provides our concluding remarks and some future work avenues.

2 Literature review

The design of sustainable and efficient supply chains in agriculture has gained increasing attention in recent years, particularly in the context of closed-loop supply chains (CLSCs) and reverse logistics (RL). This section critically reviews the existing body of literature on CLSC applications in the agricultural sector, with a focus on perishable products such as olives. Given the growing global demand for horticultural products and the pressing need to address environmental concerns (such as food waste, water scarcity, and carbon emissions) researchers have explored various models and strategies to optimize agricultural supply chains. However, despite the proliferation of studies on agricultural supply chains (ASC),

there remains a notable gap in the literature concerning integrated CLSC models that simultaneously address economic and environmental objectives, particularly in the olive industry. This review aims to contextualize the current study by identifying key contributions, methodological trends, and unresolved challenges in the field.

2.1 Advances in agricultural supply chain (ASC)

The rapid growth of ASCs, driven by rising global demand and sustainability pressures, has long made the design of resilient and efficient networks indispensable. Horticultural products present unique challenges, including quality preservation, climate sensitivity, and price volatility, that require adapted approaches. Supply chains for these products are typically classified into two categories based on their life-cycle durability: perishable (e.g., olives, citrus) and non-perishable (e.g., grains, dried fruits) (Siawsolit & Gaukler, 2021). Since 2010, numerous studies have addressed ASC and fruit supply chains (Borodin et al., 2016; Soto-Silva et al., 2016), and several mathematical models have been proposed for crops like fruits, vegetables, grapes, nuts, and berries (Zhang & Wilhelm, 2011). In general, there have been quite a few reviews published in the fields of ASC and FSC that are briefly summarized in Table 1. Recent surveys mapping this space include syntheses of artificial intelligence and machine learning in ASC (Kumari et al., 2025), and reviews of emerging digital technologies

Table 1 Overview of related review papers on ASCs and FSCs

Studies	Scope	Focus	Time horizon
Ahumada and Villalobos (2009)	FSC	Application of food models	1985–2005
Shukla and Jharkharia (2013)	ASC	Considering fresh products	1989–2009
Borodin et al. (2016)	ASC	Operations research in agriculture	1984–2014
Zhu et al. (2018)	FSC	Sustainable food supply chain	1998–2016
Behzadi et al. (2018)	ASC	Agricultural risk in supply chain	1990–2016
Dania et al. (2018)	ASC	Sustainable agri-food products	1996–2017
Kamble et al. (2020)	ASC	Sustainability in ASC	2000–2017
Sharma et al. (2020)	ASC	Machine learning techniques in ASC	Until 2019
Mehmood et al. (2021)	ASC	Circular economy transition in ASC	2015–2020
Dong (2021)	ASC	Resilient & agriculture value chains	1960–2020
Grabs and Carodenuto (2021)	FSC	Sustainability governance in FCS	2015–2020
Yadav et al. (2022)	ASC	Agri-food supply chain management	2005–2020
Van Nguyen et al. (2023)	ASC	Application of blockchain in ASC	2018–2022
Trevisan and Formentini (2023)	FSC	Food loss and waste prevention	2013–2021
Charles et al. (2023)	ASC	Artificial intelligence integration	2017–2022
Nayal et al. (2023)	ASC	Sustainable agri-food performance	Until 2020
Mwangakala et al. (2024)	ASC	Emerging digital technologies	2012–2022
Sharma et al. (2024)	FSC	Industry 4.0-enabled intelligent FSC	2013–2023
Vern et al. (2025)	FSC	Blockchain technology in agri-food	2018–2022
Kumari et al. (2025)	ASC	Artificial intelligence in ASC	2006–2022
Nie et al. (2025)	FCS	Resilient agri-food supply chains	2018–2024

for more equitable agri-food chains (Mwangakala et al., 2024). The studies reviewed by these papers and the ones cited above have shaped product-specific logistics solutions; however, comprehensive models for the olive industry remain scarce, highlighting a significant research gap.

Montenegro-Dos Santos et al. (2023) present a bi-objective mixed-integer mathematical model for rescheduling olive harvesting, aimed at maximizing olive oil production while minimizing deviations from an initial plan to control schedule instability. Their model is updated with new yield and disruption information and solved using Gurobi's weighted-sum approach. Results indicate that the model effectively balances production efficiency and plan stability, enabling flexible, near-optimal decision-making, and underscores the significance of early rescheduling and renegotiation in the olive supply chain. More recently, Urrea-Calfunir et al. (2024) introduced an MILP model for the olive oil supply chain, focusing on integrating harvesting, production, transportation, and mobile mill relocation decisions. Their study aimed to maximize profits for small producers by considering factors like concentration dynamics, harvest windows, and transportation costs. Their real case study shows that decentralized production with mobile mills increases profits by 4–5% and reduces transportation distances and costs by over 40% compared to fixed-mill setups, highlighting the benefits of mobile processing in the olive oil supply chain.

In supply chain, olives can be classified as perishable garden products. The supply chain for perishable horticultural products has attracted particular attention due to rising consumer demand for fresh, high-quality produce and the managerial complexity associated with maintaining product integrity under variable climatic conditions (Avishan et al., 2025). Over the past decade, numerous studies have examined strategies to enhance competitiveness in agri-food supply chains, emphasizing responsiveness, cost efficiency, and waste reduction (Qin & Cao, 2024; Shi et al., 2025). Early contributions, such as Ahumada and Villalobos (2009), introduced foundational models for vegetable production planning, while subsequent works expanded the scope to include procurement, transportation, facility location, and information systems integration. These efforts underscore the logistical complexity of agricultural networks, where strategic decisions (such as orchard establishment and capacity planning) carry long-term implications for cost and service performance (Starbird, 1988). More recent research has explored CLSC configurations for specific crops, such as mushrooms (Banasik et al., 2017), often adopting multi-objective formulations to balance economic and environmental outcomes.

Among the limited available ASC models adaptable for the olive industry, the following studies are noteworthy. Considering the special characteristic of virgin olive oil in terms of health benefits, Roselli et al. (2017) assessed the competitiveness of this product within a supply chain network in the present market by conducting various scenarios. Surucu-Balci and Tuna (2021) identified and ranked logistics-related food loss drivers in the fruit and vegetable supply chain, concentrating on fruits and vegetables products. Similarly, Magalhaes et al. (2021) investigated the root causes of waste in fruit and vegetables industry.

Furthermore, a multi-objective MILP model was offered by Javanmardan et al. (2024) to minimize the costs and emissions of an ASC with their model seeking to maximize responsiveness of the chain to the demand for fruit, vinegar, and compost across various markets. Their findings indicated that achieving high responsiveness in such a system would incur substantial network costs. Recently, Wang (2024) focused on perishable products and presented an integrated two-stage flow shop scheduling problem with order assignment and lead-time constraints for the fruit and vegetable supply chains. On the other hand, the model proposed by Shahsavani et al. (2024) realized a zero-waste outcome by eliminating waste in plastic crate usage and fruit residuals using a circular economy (CE) approach, decreasing

the total costs of such systems by one percent compared to linear counterparts. In recent study by Goodarzian et al. (2025), a three-echelon supply chain network was proposed for agri-food products to reduce both the costs and CO_2 emission. The authors compared multiple metaheuristic algorithms in addition to the Augmented Epsilon-constraint method. The application of water consumption, though, was not considered in their approach.

2.2 CLSCs and RL for agricultural products

Despite the growing body of research on agricultural supply chains, studies specifically addressing Closed-Loop Supply Chains (CLSCs) and Reverse Logistics (RL) for agricultural fruits remain limited (refer to the summary of the available reviews provided in Table A1, Appendix A). These foundational studies laid the groundwork for integrating sustainability into agricultural logistics, yet they often focused on isolated aspects such as energy recovery or cost minimization, without fully addressing the interconnected economic, environmental, and operational dimensions of CLSCs in fruit supply chains.

Building on these early efforts, subsequent research broadened the scope of CLSC studies to incorporate multiple dimensions, including economic, energy, and environmental objectives. For instance, Alfonso-Lizarazo et al. (2013) proposed a CLSC network for palm oil, and considered economic and energy related objectives. Liao et al. (2020) presented a practical CLSC model for citrus crates validated through a real-world case study from north of Iran, while Seydanlou et al. (2022) introduced a multi-objective framework for olives that jointly minimizes costs and greenhouse gas emissions under facility-opening constraints. The amount of product flow and inventory level along with the opened facilities are determined using their model.

Additional contributions include Goodarzian et al. (2023), who configured a sustainable citrus CLSC by integrating production, distribution, inventory control, recycling, and locational decisions. Their results suggest that citrus waste can be transformed into value-added products, supporting a sustainable bio-economy strategy. A CLSC for pistachio products was also proposed by Rajabi-Kafshgar et al. (2023), minimizing the total costs comprise the fixed opening costs, production, and transportation costs. Their proposed settings along with Taguchi experimental design method claimed to reduce the total costs for pistachio products.

2.3 Research opportunities and contributions

Despite the growing body of work on olive supply chains, several critical gaps remain unaddressed. Most existing models focus solely on forward logistics, neglecting the integration of reverse flows, packaging recovery, and wastewater management. This study addresses these limitations by proposing a unified closed-loop supply chain (CLSC) framework that incorporates both forward and reverse logistics, explicitly modeling packaging reuse and the circulation of wastewater and rich-water streams. Furthermore, prior models typically optimize either economic or environmental objectives in isolation. In contrast, our approach adopts a dual-objective formulation that simultaneously minimizes total cost and total water consumption, thereby capturing the trade-offs between profitability and sustainability. Unlike previous studies that treat olives as a homogeneous product, our model differentiates between green, black, and degraded olives, assigning each to distinct destinations such as table consumption, oil extraction, fertilizer production, or animal feed.

Another novel contribution lies in the operationalization of packaging circularity. We incorporate package defect rates, reuse potential, water requirements for recovery, and associated costs directly into the model. Water accounting is extended beyond processing stages to include irrigation, enabling a comprehensive assessment of water use across the entire supply chain. We also analyze how variations in water collection efficiency and loss rates affect overall supply chain performance—an aspect largely overlooked in existing literature. Additionally, the model captures the propagation of post-harvest degradation rates through cost structures, circularity metrics, and water footprints. Systematic scenario analyses are conducted to evaluate these effects. Importantly, our findings are translated into actionable managerial insights, linking model outputs to decisions on cultivation practices, procurement strategies, packaging selection, recovery system design, and integrated water management.

3 Problem statement and system description

To achieve sustainable production and efficient water use across olive cultivation, harvesting, and processing stages, this study proposes a comprehensive CLSC model. The proposed model aims to minimize the cost while minimizing the water consumption of harvesting, processing, and cleaning. Here we need to capture three primary material flows: (1) olive fruits and their derived products from gardens to demand centers, (2) packaging materials from production to recovery and reuse, and (3) water flows across irrigation, processing, and recovery stages. Therefore, the proposed model addresses the environmental impact of water consumption, and pollution, and package waste by enabling their circulation and repurposing within the use cycle, while also minimizing costs. The system structure is illustrated in Fig. 2, which outlines the interconnected flows and facility types within the network. The.

3.1 Olive product flow

In the creation of a CE network for olive production and consumption, it is essential to identify the inputs and outputs of each processing center to match the proper material flow. The total demand types for olive fruit are green table olive and olive oil, and each of them has different harvesting times, processing types, water consumption, and packaging. Green olives are harvested in September or early October (commonly referred to as early harvest), while purple or black olives are harvested during autumn or early winter, typically from October to December, which is called main or late harvest. The green harvested olives are transported to processing plants in Polyvinyl chloride (PVC) or High-Density Polyethylene (HDPE) crates to be cleaned and washed (to remove dust, dirt, and leaves), sorted and graded (to separate degraded olives from healthy olives), cured and fermented (to debitter olives) in Spanish style, natural style, or salt water, and stabilized (pasteurized or preserved in brine or acid) (Spada et al., 2025). The green table olives are packaged in glass jars, stainless steel tins, or Polyethylene (PE) film pouches and then distributed to demand spots, including retailers, restaurants, and food industries.

To produce olive oils, mainly purple and black olives are required to be harvested and transported to the mill in addition to degraded green olives. The olives delivered in bins to mills are cleaned and washed (for removal of dust, dirt, and leaves), milled or crushed into paste, malaxated (slow mixing of paste, usually 20–40 min, controls oil droplet coalescence, temperature affects phenolics and aroma), oil is separated from water and pomace. The produced olive oil undergoes acidity level testing to classify it into: (a) extra virgin olive oil

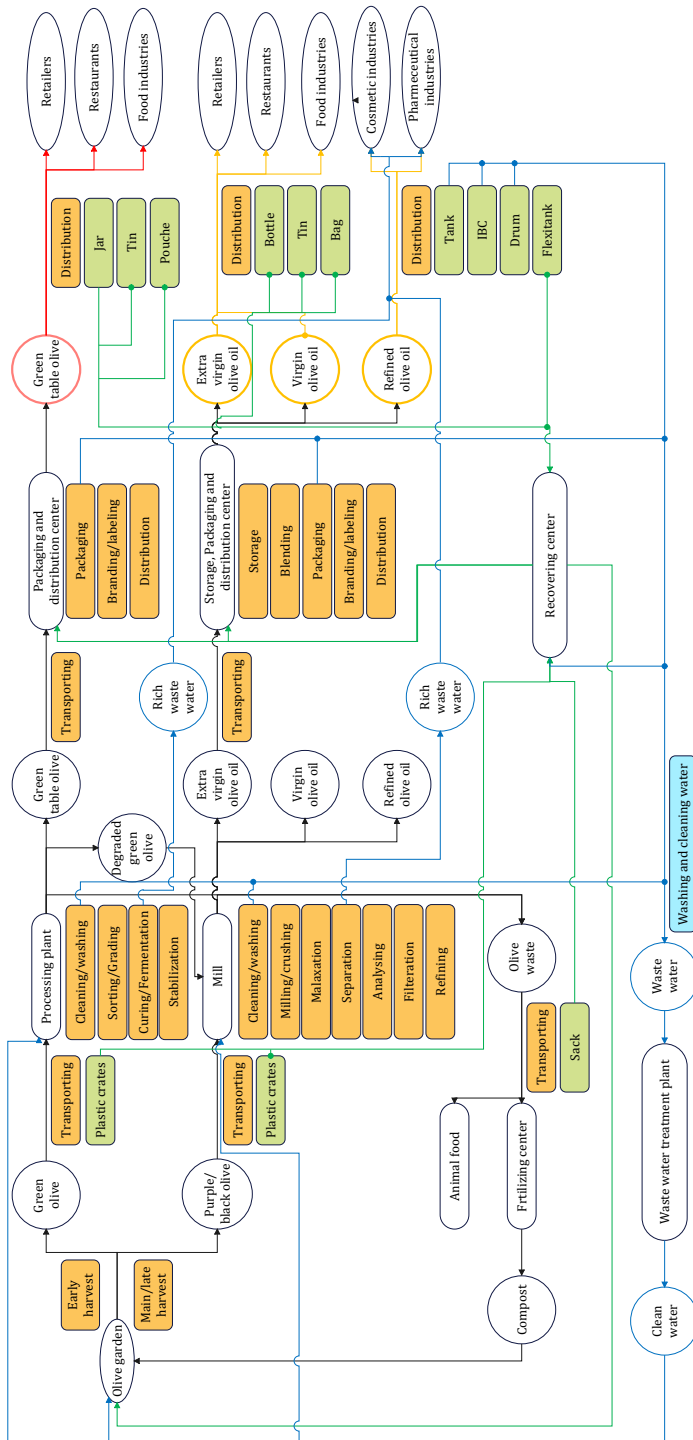


Fig. 2 The schematic process flow of the studied supply chain for model development

(less than 0.8% acidity and no defects), (b) virgin olive oil (less than 2% acidity and slight defects), and (c) lampante (with too high acidity or serious sensory defects) which is inedible and requires a mechanical or chemical refining process to yield refined olive oil (Seifollahi et al., 2024). Extra virgin and virgin olive oils are edible and thus they are transported to retailers, restaurants, and food industries in glass, polyethylene terephthalate (PET), and HDPE bottles, stainless steel tins, or in PET and HDPE bags. On the other hand, the refined olive oil is assigned to cosmetic and pharmaceutical industries in ISO tanks, Intermediate bulk containers (IBC), drums, and flex tanks. In addition to the possibility of oiling from the degraded green fruits, they can also be sent to either fertilizer manufacturing centers to produce compost (and subsequently returned to gardens) or the animal food industry.

3.2 Packaging flow, reusing, and recovery

To mitigate environmental burden from package waste in distribution from early stages to consumption points, the proposed model aims to collect packages and recover them into use cycles through recyclable packages and reusing them. In this case, three phases of collection are contemplated, including collecting packages such as crates, bins, and sacks from processing centers (processing plants, mills, fertilizer manufacturing centers, and the animal food industry); collecting packages from consumption points (retailers, restaurants, food industries, cosmetic industries, and pharmaceutical industries); and collecting and recovering containers for reuse from pharmaceutical and cosmetic industries after transporting refined olive oil to them. The collected packages can be reused after washing and cleaning; otherwise, they should be recycled and reintroduced to the use cycle. By adopting this method, not only is the environmental damage from package waste mitigated, but also the reliance on virgin resources is reduced in aim of achieving CE principles.

3.3 Water flows and recovery

To address the water crisis, recoverable water waste from every point of the olive network should be traced and treated to return clarified water to the use cycle for irrigating or washing purposes. In this regard, washing and cleaning green, purple, and black olives' water (wastewater); curing green olive water process (rich wastewater); oiling process water (rich wastewater); recovering packages process water from washing packages with detergent and hot water (wastewater); and washing olive oil containers with detergent and hot water (wastewater) are identified as traceable water that should be collected and treated. It must be noted that rich wastewater from processing plants and mills cannot be reused directly for either irrigation or washing purposes due to its content. Water from curing green olive processing contains large volumes of brine or lye solutions that can be valuable input for cosmetic and pharmaceutical purposes after valorization. Additionally, the rich water waste from the oiling process contains phenolics and polyphenols that can be extracted through valorization and subsequently fed to cosmetic and pharmaceutical companies (Donner & de Vries, 2021).

In this way, not only can valuable material be reintroduced to the use and revenue generation cycle, but also the environmental concern regarding water waste is addressed. However, collected water waste must be transported to a wastewater treatment plant for treatment and clarification. The clarified water can then be reused for either washing and cleaning processes in processing plants and mills or olive gardens for irrigation.

3.4 Olive fruit categories

Olive fruits can be categorized into green, purple, black, and degraded olives according to harvest time and quality that can affect water consumption, procurement volume, demand point, cost, as well as the required processing stages. As mentioned, green olives are harvested early in September and early October to fulfill demand for green table olives that are not suitable for producing olive oil. However, degraded green olives can produce olive oil but are not comparable to purple and black olives, which are riper. In this case, degraded green olives can be sent either to a mill for oiling processes or to fertilizer production centers and animal feed industries for downcycling. On the other hand, purple and black olives are harvested from October to December for oiling purposes since they are ripe and generate more olive oil.

3.5 Model assumptions

The following assumptions underpin the studied system:

- The quality of harvested olives varies due to environmental conditions such as climate, soil condition, and pest pressure, and is represented in the model by the parameters α_1 and α_2 for early and main harvest stages.
- The quality parameters (i.e., α_1 and α_2) directly influence the volume of olives entering processing plants and mills; i.e., higher α values indicate better crop quality, resulting in lower degradation rates and improved suitability for table olive or oil production.
- A static single-period decision horizon is adopted, and time-dependent scheduling decisions are not considered.
- The irrigation water is unrecoverable.
- Rich wastewater requires treatment or valorization before reuse that must be collected and repurposed in the pharmaceutical and cosmetic industries.
- Water waste and rich wastewater are recoverable, subject to a predefined collection loss rate.
- Package quality is considered. The higher the quality of packages, the higher the reuse rate. Parameter *def* explicitly represents the defective package rate, indicating the recovery volume.
- Facility setup costs are fixed.
- The market demand is deterministic in the base case.
- Transportation cost is distance-based and proportional to volume.

It is worth noting that, in addition to the above direct effect considered for the quality parameters, they also influence downstream cost, water use, and waste generation.

4 Mathematical model and uncertainty framework

In this section, a bi-objective model is formulated to minimize the total cost and overall water consumption under three material streams. The sets, parameters, and variables of the proposed model are defined in Table 2.

Table 2 Notations of the proposed model

Sets	Definition
G	Set of olive gardens, indexed by g
O	Set of processing plants, indexed by o
M	Set of mills, indexed by m
D	Set of packaging and distribution centers, indexed by d
R	Set of recycling centers, indexed by r
T	Set of wastewater treatment plants, indexed by t
S	Set of retailers, indexed by s
H	Set of restaurants (hospitality centers), indexed by h
F	Set of food industries, indexed by f
C	Set of cosmetic industries, indexed by c
E	Set of pharmaceutical industries, indexed by e
K	Set of fertilizer Manufacturing centers, indexed by k
A	Set of animal food, indexed by a
P	Set of products, indexed by p ; $P = \{1, 2, \dots, P \}$ (see Table A3 for the complete list)
P'_i	A family of product type subsets, where i denotes the index defined in Table A4 ($P'_i \subset P$)
N	Set of packages, indexed by n ; $N = \{1, 2, \dots, N \}$ (see Table A5 for the complete list)
N'_j	A family of package type subsets, where j denotes the index defined in Table A6 ($N'_j \subset N$)
W	Set of water type, indexed by w ; $W = \{1, 2, \dots, W \}$ (see Table A7 for the complete list)
Parameters	
$dm_{i,p,n}$	The demand of market $i \in S \cup H \cup F \cup C \cup E$ for product $p \in P$ in package $n \in N$
$pcog,p$	Purchasing cost of olive fruit p from garden $g \in G$
pcn_n	Procuring cost of one unit package $n \in N$
pcw_w	Procuring cost of a liter of water type $w \in W$
ebp_p	Economic (fiscal) benefit of repurposing product $p \in P$
sci	Set-up cost of facility $i \in O \cup M \cup D \cup R \cup T \cup K \cup A$
cwp_o	Cost of Washing and cleaning process in Processing plant $o \in O$ per kilogram olive
cgp_o	Cost of Grading and sorting process in Processing plant $o \in O$ per kilogram olive
ccp_o	Cost of Curing process in Processing plant $o \in O$ per kilogram olive
csp_o	Cost of Stabilization process in Processing plant $o \in O$ per kilogram olive
cwm_m	Cost of Washing and cleaning process in Mill $m \in M$ per kilogram olive
ccm_m	Cost of Crushing and milling process in Mill $m \in M$ per kilogram olive
cmm_m	Cost of Malaxation process in Mill $m \in M$ per kilogram olive
csm_m	Cost of Separation process in Mill $m \in M$ per kilogram olive
cam_m	Cost of Analyzing process in Mill $m \in M$ per kilogram olive
cfm_m	Cost of Filtration process in Mill $m \in M$ per kilogram olive
crm_m	Cost of Refining process in Mill $m \in M$ per kilogram olive
$cod_{d,p,n}$	Cost of Operations in Distribution center $d \in D$ per unit of product type $p \in P$ in package type $n \in N$

Table 2 (continued)

Sets	Definition
crp_n	Cost of Recovering process of Package $n \in N$
cdf_k	Cost of Degraded olive fruit in Fertilizers center $k \in K$
caf_a	Cost of Animal Feed process of degraded olive fruit in recovering center $a \in A$
cww_t	Cost of Wastewater clarification process in wastewater treatment plant $t \in T$
$hcp_{i,p}$	Holding (Storage) cost of product $p \in P$ in center $i \in O \cup M \cup D$
$tcp_{i,j,p,n}$	Transportation cost of product $p \in P$ from location i to location j in package $n \in N$; $i, j \in G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A$
$cp_{g,p}$	Capacity of garden $g \in G$ for harvesting crop $p \in P_1'$
$cpc_{i,p}$	Capacity of center $i \in O \cup M$ for processing product type $p \in P$
$cpd_{d,p,n}$	Capacity of distribution center $d \in D$ of product $p \in P_2'$ in package $n \in N$
$cpr_{r,n}$	Capacity of recycling center $r \in R$ for package $n \in N$
cpt_t	Capacity of water treatment plant $t \in T$
cpf_k	Capacity of fertilizer manufacturing center $k \in K$
cpa_a	Capacity of animal feed facility $a \in A$
cpp_n	Capacity of package type $n \in N$ for transporting products
ir_p	The required irrigation (liters of water) for growing a kilogram of product $p \in P_3'$
wop_p	The required liters of water to wash a kilogram of olive product type $p \in P_1'$
wp_n	The required liters of water to wash a package of type $n \in N$
wc	The required liters of water for curing green olive process
ws	The required liters of water for the separation stage in olive oil extraction process
lw	Loss portion of water collection after washing olive crops
lc	Loss portion of water collection after curing green olive process
ls	Loss portion of water collection after separation olive oil process
lp	Loss portion of water collection after washing packages
α_1	Quality rate of harvested green olives
α_2	Quality rate of harvested purple/black olives
β_1	Portion of extra virgin olive oil produced after oiling process
β_2	Portion of virgin olive oil produced after oiling process
β_3	Portion of refined olive oil produced after oiling process
def_n	Defective rate of Package type $n \in N$ in the transportation process
Variables	
$QT_{i,j,p,n}$	The quantity of shipped product $p \in P$ from spot i to spot j in package $n \in N$; $i, j \in G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A$
$QP_{i,p}$	The quantity of processed product $p \in P$ in center i ; $i \in O \cup M \cup D \cup K \cup A$
$QH_{i,p}$	The quantity of stored product $p \in P$ in center $i \in O \cup M \cup D$
$QN_{i,j,p,n}$	The quantity of required package $n \in N$ to transport product $p \in P$ from spot i to spot j ; $i, j \in G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A$

Table 2 (continued)

Sets	Definition
$QR_{i,j,p,n}$	The quantity of package $n \in N$ to transport product $p \in P$ from center i to j are defected and must be recycled; $i, j \in G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A$
$QW_{i,w}$	The quantity of consumed/produced water $w \in W$ in center $i \in G \cup O \cup M \cup D \cup R$
X_i	Activation of center $i \in O \cup M \cup D \cup R \cup T \cup K \cup A$. 1 if center i is opened; 0 otherwise

4.1 Total cost objective function

The first objective function, as given in Eq. (1), minimizes the total cost of a centralized olive supply chain network, including: the cost of procuring green, purple, and black olives, packages, and water; the cost of setting up a processing facility (processing plant, mill, packaging and distribution, recovering center, wastewater treatment plant, fertilizer center, and animal food industry); the cost of processing in each opened facility; the cost of transportation from one location to another; the cost of storage; and finally, subtracting the gained revenue from selling the fertilizers, animal food, and rich wastewater. To present this objective function compactly, let $V = G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A$ denote the set of network nodes and let $\mathcal{A} \subset V \times V$ denote the set of admissible transportation arcs. For each arc $(i, j) \in \mathcal{A}$, the set $\Phi_{i,j} \subseteq P \times N$ contains the admissible product–package pairs (p, n) that may flow on that arc, as enumerated in Table A8 (Appendix A). Furthermore, the forward arcs are represented by $\mathcal{A}' = \mathcal{A} \setminus (K \times G)$, distinguishing them from the reverse package–return arcs $K \times G$.

$$\min Z_1 = C^{\text{proc}} + C^{\text{pack}} + C^{\text{water}} + C^{\text{setup}} + C^{\text{oper}} + C^{\text{recov}} + C^{\text{inv}} + C^{\text{trans}} - B^{\text{rev}} \quad (1)$$

where its components respectively capture the olive and package procurement, water acquisition, facility setup, processing and distribution operations, package recovery, storage, transportation, and the revenues gained from repurposing fertilizer, animal feed, and rich wastewater. These components are defined mathematically as:

(a) Olive fruit procurement (gardens to processing plants and mills):

$$C^{\text{proc}} = \sum_{g \in G} \sum_{j : (g, j) \in \mathcal{A}} \sum_{(p,n) \in \Phi_{g,j}} QT_{g,j,p,n} \cdot pcog,p$$

(b) Package procurement (all arcs, priced on the non-defective fraction):

$$C^{\text{pack}} = \sum_{(i,j) \in \mathcal{A}} \sum_{(p,n) \in \Phi_{i,j}} QN_{i,j,p,n} \cdot (1 - def_n) \cdot pcn_n$$

(c) Water procurement:

$$C^{\text{water}} = \sum_{i \in G \cup O \cup M \cup D \cup R} \sum_{w \in W} QW_{i,w} \cdot pcw_w$$

(d) Facility setup:

$$C^{\text{setup}} = \sum_{i \in O \cup M \cup D \cup R \cup T \cup K \cup A} X_i \cdot sc_i$$

(e) Processing and operational costs. Let $\varphi_{i,p}$ denote the aggregate processing cost (per unit) of product p at facility i , and Π_i the set of products processed at facility i (both defined in Table A9, Appendix A):

$$C^{\text{oper}} = \sum_{i \in O \cup M \cup K \cup A} \sum_{p \in \Pi_i} QP_{i,p} \cdot \varphi_{i,p} + \sum_{t \in T} QW_{t,4} \cdot cw_{w_t} \\ + \sum_{d \in D} \sum_{j : (d,j) \in A} \sum_{(p,n) \in \Phi_{d,j}} QT_{d,j,p,n} \cdot cod_{d,p,n}$$

(f) Package recovery. Defective packages on forward arcs are recycled, whilst returnable sacks are recovered on the non-defective fraction using the reverse arcs (i.e., $K \rightarrow G$):

$$C^{\text{reco}} = \sum_{(i,j) \in \mathcal{A}'} \sum_{(p,n) \in \Phi_{i,j}} QN_{i,j,p,n} \cdot def_n \cdot cr_{p_n} + \sum_{k \in K} \sum_{g \in G} QN_{k,g,9,4} \cdot (1 - def_4) \cdot cr_{p_4}$$

(g) Inventory holding. Let Π_i' denote the storable products at node i , with $\Pi'_0 = \{5\}$, $\Pi'_m = P_4'$, $\Pi'_d = P_2'$:

$$C^{\text{inv}} = \sum_{i \in O \cup M \cup D} \sum_{p \in \Pi'_i} QH_{i,p} \cdot hcp_{i,p}$$

(h) Transportation:

$$C^{\text{trans}} = \sum_{(i,j) \in \mathcal{A}} \sum_{(p,n) \in \Phi_{i,j}} QT_{i,j,p,n} \cdot tcp_{i,j,p,n} + \sum_{a \in A} QP_{a,10} \cdot tcp_{a,10,4}$$

(i) Repurposing revenues (fertilizer, animal feed, and rich wastewater):

$$B^{\text{rev}} = \sum_{i \in K \cup A} \sum_{p \in \Pi_i} QP_{i,p} \cdot eb_p + \sum_{i \in O \cup M} QW_{i,3} \cdot eb_3$$

4.2 Water consumption objective function

The second objective function (Z_2) minimizes the water consumption of harvesting olives, processing olives into green table olives and olive oils, packaging olive products, and recovering packages processed by reintroducing clarified wastewater in the olive supply chain network, as shown in Eq. (2).

$$\min Z_2 = \min(\text{WaterConsumption}) = \sum_{i \in G \cup O \cup M \cup D \cup R} (QW_{i,1} - QW_{i,4}) \quad (2)$$

4.3 Constraints

Let α_1 denote the quality of harvested green olives that indicates the acceptable quality of harvested olives that can be utilized in green table olive production. Low-quality green olives are identified as degraded green olives that can be transported to mills for olive oil production or fertilizer manufacturing centers and animal food industries. Additionally, variable QH acts as a surplus balancing term, capacity buffer, and flow slack variables to maintain consistency between input and output of each center, indicating momentary storage (storage capacity). In other words, storage in this model reflects short-term balancing capacity within

a static strategic framework, not dynamic inventory accumulation over multiple time periods. Equations (3) and (4) calculate the volume of produced green table olives and degraded olives after the sorting and grading process, respectively.

$$\sum_{g \in G} QT_{g,o,1,1} \times \alpha_1 = QP_{o,5} = QH_{o,5} + \sum_{d \in D} \sum_{n \in N'_1} QT_{o,d,5,n} \quad \forall o \in O \quad (3)$$

$$\sum_{g \in G} QT_{g,o,1,1} \times (1 - \alpha_1) = QP_{o,3} = \sum_{m \in M} QT_{o,m,3,1} + \sum_{k \in K} QT_{o,k,3,1} + \sum_{a \in A} QT_{o,a,3,1} \quad \forall o \in O \quad (4)$$

Similar to the harvested green olives, quality of harvested purple and black olives also matters in olive oil production. Parameter α_2 is assumed to be the quality of all inputs to the mill (i.e., degraded green olives and harvested purple and black olives). Equations (5)–(7) calculate the produced extra virgin olive oil, virgin olive oil, and refined olive oil (which involves a refining process), respectively.

$$\left(\sum_{g \in G} QT_{g,m,2,1} + \sum_{o \in O} QT_{o,m,3,1} \right) \times \alpha_2 \beta_1 = QP_{m,6} = QH_{m,6} + \sum_{d \in D} \sum_{n \in N'_2} QT_{m,d,6,n} \quad \forall m \in M \quad (5)$$

$$\left(\sum_{g \in G} QT_{g,m,2,1} + \sum_{o \in O} QT_{o,m,3,1} \right) \times \alpha_2 \beta_2 = QP_{m,7} = QH_{m,7} + \sum_{d \in D} \sum_{n \in N'_2} QT_{m,d,7,n} \quad \forall m \in M \quad (6)$$

$$\left(\sum_{g \in G} QT_{g,m,2,1} + \sum_{o \in O} QT_{o,m,3,1} \right) \times \alpha_2 \beta_3 = QP_{m,8} = QH_{m,8} + \sum_{d \in D} \sum_{n \in N'_2} QT_{m,d,8,n} \quad \forall m \in M \quad (7)$$

Low-quality inputs to mills are not suitable to produce olive oil, which is separated and allocated to fertilizer manufacturing centers and animal food industries to not only increase the resource value but also reduce landfilling, as stated in Eq. (8).

$$\left(\sum_{g \in G} QT_{g,m,2,1} + \sum_{o \in O} QT_{o,m,3,1} \right) \times (1 - \alpha_2) = QP_{m,4} = \sum_{i \in K \cup A} QT_{m,i,4,1} \quad \forall m \in M \quad (8)$$

Cured green table olives from processing plants and produced olive oils from mills must be packaged, labeled, and distributed to demand points, including retailers, restaurants, food industries, cosmetic industries, and pharmaceutical industries, while considering the storage capacity for excess, as shown in Eqs. (9)–(12).

$$\sum_{o \in O} \sum_{n \in N'_1} QT_{o,d,5,n} = QH_{d,5} + \sum_{i \in S \cup H \cup F} \sum_{n \in N'_3} QT_{d,i,5,n} \quad \forall d \in D \quad (9)$$

$$\sum_{m \in M} \sum_{n \in N'_2} QT_{m,d,6,n} = QH_{d,6} + \sum_{i \in S \cup H \cup F} \sum_{n \in N'_4} QT_{d,i,6,n} \quad \forall d \in D \quad (10)$$

$$\sum_{m \in M} \sum_{n \in N'_2} QT_{m,d,7,n} = QH_{d,7} + \sum_{i \in S \cup H \cup F} \sum_{n \in N'_4} QT_{d,i,7,n} \quad \forall d \in D \quad (11)$$

$$\sum_{m \in M} \sum_{n \in N'_2} QT_{m,d,8,n} = QH_{d,8} + \sum_{i \in C \cup E} \sum_{n \in N'_5} QT_{d,i,8,n} \quad \forall d \in D \quad (12)$$

Equations (13) and (14) indicate the inputs, operated, and output of each fertilizer center and animal food industry.

$$\sum_{o \in O} QT_{o,k,3,1} + \sum_{m \in M} QT_{m,k,4,1} = QP_{k,9} = \sum_{g \in G} QT_{k,g,9,4} \quad \forall k \in K \quad (13)$$

$$\sum_{o \in O} QT_{o,a,3,1} + \sum_{m \in M} QT_{m,a,4,1} = QP_{a,10} \quad \forall a \in A \quad (14)$$

Let cpp_n denotes the volume of material that can be carried by each package $n \in N$. Equation (15) measures the quantity of required packages to transport a volume of products from one location to another.

$$QN_{i,j,p,n} = \left\lceil QT_{i,j,p,n} / cpp_n \right\rceil \quad \forall p \in P, n \in N, \\ \forall i, j \in G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A \quad (15)$$

Linearization: Eq. (15) calculates the minimum integer number of packages (ceiling function). Given that this value is naturally minimized by the first objective function, Eq. (15) can technically be replaced with the following linear equivalent:

$$QN_{i,j,p,n} \geq \left(QT_{i,j,p,n} / cpp_n \right) \quad \forall p \in P, n \in N, \\ \forall i, j \in G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A \quad (16)$$

In every transportation process, packages may be damaged or contaminated, they must undergo washing, cleaning, and recovery processes so that they can be reintroduced to the use cycle. By introducing the parameter, the number of packages that must be collected and recovered is determined in Eq. (17). This allows us to reduce the environmental impact and increase the resource value under circular economic principles.

$$QR_{i,j,p,n} = QN_{i,j,p,n} \times def_n \quad \forall p \in P, n \in N, \\ \forall i, j \in G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A. \quad (17)$$

The second objective of the study aims to minimize the water consumption through water recovery as circular economy principles. Therefore, generated wastewater from processes in the olive supply chain must be clarified and returned to the use cycle. Equations (18)–(22) are respectively formulated to calculate water used in olive gardens from fresh (fw) and clarified (cw) water resources for: irrigation purposes; washing olive fruits, curing process water, washing packages in processing plants; washing olive fruits, separating process water, washing packages in mills; washing packages in packaging and distribution centers; and washing packages in recovery centers.

$$QW_{g,1} + QW_{g,4} \geq \left(\sum_{o \in O} QT_{g,o,1,1} / ir_1 \right) + \left(\sum_{m \in M} QT_{g,m,2,1} / ir_2 \right) \quad \forall g \in G \quad (18)$$

$$QW_{o,1} + QW_{o,4} \geq \left(\sum_{g \in G} QT_{g,o,1,1} / wo_1 \right) + \left(\sum_{g \in G} QT_{g,o,1,1} \times \alpha_1 / wc \right) \\ + \left(\sum_{d \in D} \sum_{n \in N'_1} QN_{o,d,5,n} / wp_n \right) \quad \forall o \in O \quad (19)$$

$$QW_{m,1} + QW_{m,4} \geq \left(\sum_{g \in G} QT_{g,m,2,1} / wo_2 \right) + \left(\sum_{o \in O} QT_{o,m,3,1} / wo_3 \right) \\ + \left(\sum_{g \in G} QT_{g,m,2,1} \times \alpha_2 / ws \right) + \left(\sum_{o \in O} QT_{o,m,3,1} \times \alpha_2 / ws \right) \\ + \left(\sum_{d \in D} \sum_{p \in P'_4} \sum_{n \in N'_2} QN_{m,d,p,n} / wp_n \right) \quad \forall m \in M \quad (20)$$

$$\begin{aligned}
QW_{d,1} + QW_{d,4} \geq & \left(\sum_{j \in S \cup H \cup F} \sum_{n \in N'_3} QN_{d,j,5,n} / wp_n \right) \\
& + \left(\sum_{j \in S \cup H \cup F} \sum_{p \in P'_5} \sum_{n \in N'_4} QN_{d,j,5,n} / wp_n \right) \\
& + \left(\sum_{j \in C \cup E} \sum_{n \in N'_5} QN_{d,j,8,n} / wp_n \right) \quad \forall d \in D \quad (21)
\end{aligned}$$

$$\begin{aligned}
QW_{r,1} + QW_{r,4} \geq & \left(\sum_i \sum_j \sum_{p \in P} \sum_{n \in N} QR_{i,j,p,n} \times def_n / wp_n \right) \quad \forall r \in R, \\
& \forall i, j \in G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A \quad (22)
\end{aligned}$$

Equations (23)–(26) calculate the wastewater (ww) of processes in processing plants, mills, packaging and distribution centers, and recovery centers by applying water collection's loss portion, respectively.

$$\begin{aligned}
QW_{o,2} = & \left(\sum_{g \in G} QT_{g,o,1,1} / wo_1 \right) \times (1 - lw) \\
& + \left(\sum_{d \in D} \sum_{n \in N'_1} QN_{o,d,5,n} / wp_n \right) \times (1 - lp) \quad \forall o \in O. \quad (23)
\end{aligned}$$

$$\begin{aligned}
QW_{m,2} = & \left(\sum_{g \in G} QT_{g,m,2,1} / wo_2 \right) \times (1 - lw) \\
& + \left(\sum_{o \in O} QT_{o,m,3,1} / wo_3 \right) \times (1 - lw) \\
& + \left(\sum_{d \in D} \sum_{p \in P'_4} \sum_{n \in N'_2} QN_{m,d,p,n} / wp_n \right) \times (1 - lp) \quad \forall m \in M. \quad (24)
\end{aligned}$$

$$\begin{aligned}
QW_{d,2} = & \left(\sum_{j \in S \cup H \cup F} \sum_{n \in N'_3} QN_{d,j,5,n} / wp_n \right) \times (1 - lp) \\
& + \left(\sum_{j \in S \cup H \cup F} \sum_{p \in P'_5} \sum_{n \in N'_4} QN_{d,j,5,n} / wp_n \right) \times (1 - lp) \\
& + \left(\sum_{j \in C \cup E} \sum_{n \in N'_5} QN_{d,j,8,n} / wp_n \right) \times (1 - lp) \quad \forall d \in D \quad (25)
\end{aligned}$$

$$\begin{aligned}
QW_{r,2} = & \left(\sum_i \sum_j \sum_{p \in P} \sum_{n \in N} QR_{i,j,p,n} \times def_n / wp_n \right) \times (1 - lp) \quad \forall r \in R, \\
& \forall i, j \in G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A \quad (26)
\end{aligned}$$

As mentioned, rich wastewater (*rw*) can be directly used for irrigation or processing purposes and contains large volumes of valuable brine, lye, phenolics, and polyphenols that can be reused for cosmetic or pharmaceutical purposes. The volumes of rich wastewater from curing processes in each processing plant and olive oil extraction in each mill are formulated in Eqs. (27)–(28) with consideration of the collection's loss portion that must be collected and transported to cosmetic and pharmaceutical industries.

$$QW_{o,3} = \left(\sum_{g \in G} QT_{g,o,1,1} \times \alpha_1 / wc \right) \times (1 - lc) \quad \forall o \in O \quad (27)$$

$$\begin{aligned}
QW_{m,3} = & \left(\sum_{g \in G} QT_{g,m,2,1} \times \alpha_2 / ws \right) \times (1 - ls) \\
& + \left(\sum_{o \in O} QT_{o,m,3,1} \times \alpha_2 / ws \right) \times (1 - ls) \quad \forall m \in M \quad (28)
\end{aligned}$$

Equation (29) balances the input and output of each wastewater treatment plant, leading to clarified water that can be repurposed for irrigation and washing.

$$\sum_{i \in O \cup M \cup D \cup R} QW_{i,2} = QW_{t,4} = \sum_{i \in G \cup O \cup M \cup D \cup R} QW_{i,4} \quad \forall t \in T \quad (29)$$

Equations (30)–(37) ensure that the demand for olive products, including green table olives, extra virgin olive oil, virgin olive oil, and refined olive oil, in each retailer, restaurant, food industry, cosmetic industry, and pharmaceutical industry is satisfied without any shortage.

$$\sum_{d \in D} QT_{d,s,5,n} \geq dm_{s,5,n} \quad \forall s \in S, n \in N'_3 \quad (30)$$

$$\sum_{d \in D} QT_{d,s,p,n} \geq dm_{s,p,n} \quad \forall s \in S, p \in P'_5, n \in N'_4 \quad (31)$$

$$\sum_{d \in D} QT_{d,h,5,n} \geq dm_{h,5,n} \quad \forall h \in H, n \in N'_3 \quad (32)$$

$$\sum_{d \in D} QT_{d,h,p,n} \geq dm_{h,p,n} \quad \forall h \in H, p \in P'_5, n \in N'_4 \quad (33)$$

$$\sum_{d \in D} QT_{d,f,5,n} \geq dm_{f,5,n} \quad \forall f \in F, n \in N'_3 \quad (34)$$

$$\sum_{d \in D} QT_{d,f,p,n} \geq dm_{f,p,n} \quad \forall f \in F, p \in P'_5, n \in N'_4 \quad (35)$$

$$\sum_{d \in D} QT_{d,c,8,n} \geq dm_{c,8,n} \quad \forall c \in C, n \in N'_5 \quad (36)$$

$$\sum_{d \in D} QT_{d,e,8,n} \geq dm_{e,8,n} \quad \forall e \in E, n \in N'_5 \quad (37)$$

Equations (39)–(54) guarantee that the volume of harvested, processed, and clarified materials in each garden, processing plant, mill, packaging and distribution center, recovery center, fertilizer center, animal food industry, and wastewater treatment plant does not surpass their capacity, considering the setup variables.

$$\sum_{o \in O} QT_{g,o,1,1} \leq cpg_{g,1} \quad \forall g \in G \quad (39)$$

$$\sum_{m \in M} QT_{g,m,2,1} \leq cpg_{g,2} \quad \forall g \in G \quad (40)$$

$$\sum_{g \in G} QT_{g,o,1,1} \leq cpc_o \times X_o \quad \forall o \in O \quad (41)$$

$$\sum_{g \in G} QT_{g,m,2,1} \leq cpc_m \times X_m \quad \forall m \in M \quad (42)$$

$$\sum_{s \in S} QT_{d,s,5,n} \leq cpd_{d,5,n} \times X_d \quad \forall d \in D, n \in N'_3 \quad (43)$$

$$\sum_{s \in S} QT_{d,s,p,n} \leq cpd_{d,p,n} \times X_d \quad \forall d \in D, p \in P'_5, n \in N'_4 \quad (44)$$

$$\sum_{h \in H} QT_{d,h,5,n} \leq cpd_{d,5,n} \times X_d \quad \forall d \in D, n \in N'_3 \quad (45)$$

$$\sum_{h \in H} QT_{d,h,p,n} \leq cpd_{d,p,n} \times X_d \quad \forall d \in D, p \in P'_5, n \in N'_4 \quad (46)$$

$$\sum_{f \in F} QT_{d,f,5,n} \leq cpd_{d,5,n} \times X_d \quad \forall d \in D, n \in N'_3 \quad (47)$$

$$\sum_{f \in F} QT_{d,f,p,n} \leq cpd_{d,p,n} \times X_d \quad \forall d \in D, p \in P'_5, n \in N'_4 \quad (48)$$

$$\sum_{c \in C} QT_{d,c,8,n} \leq cpd_{d,8,n} \times X_d \quad \forall d \in D, n \in N'_5 \quad (49)$$

$$\sum_{e \in E} QT_{d,e,8,n} \leq cpd_{d,8,n} \times X_d \quad \forall d \in D, n \in N'_5 \quad (50)$$

$$\sum_{o \in O} QT_{o,k,3,1} + \sum_{m \in M} QT_{m,k,4,1} \leq cpf_k \times X_k \quad \forall k \in K \quad (51)$$

$$\sum_{o \in O} QT_{o,a,3,1} + \sum_{m \in M} QT_{m,a,4,1} \leq cpa_a \times X_a \quad \forall a \in A \quad (52)$$

$$\sum_i \sum_j \sum_{p \in P} QR_{i,j,p,n} \times def_n \leq cpr_{r,n} \times X_r \quad \forall r \in R, n \in N, \\ \forall i, j \in G \cup O \cup M \cup D \cup S \cup H \cup F \cup C \cup E \cup K \cup A \quad (53)$$

$$\sum_{i \in OUMUDUR} QW_{i,2} \leq cpt_t \times X_t \quad \forall t \in T \quad (54)$$

4.4 Uncertainty framework

In this study, uncertainty originates from variability in olive quality across early and main harvest stages, represented by the parameters α_1 and α_2 , which are affected by climate conditions, pest pressure, and orchard-level agronomic factors. Because these quality parameters influence multiple downstream processes (including sorting, oil extraction percentages, degraded-fruit flows, and associated water requirements) the model must incorporate their uncertain nature. To do so, we employ a possibilistic programming framework coupled with chance-constrained linearization to retain MILP tractability while reflecting the knowledge type typically available in agri-food systems (expert judgment, sparse and heterogeneous historical records, and season-specific variability). This choice is deliberate: stochastic programming presumes well-specified probability laws and often requires large scenario trees that can be computationally onerous for mixed-integer network design; robust optimization avoids distributional assumptions but can be overly conservative. Possibilistic modeling offers a middle ground when decision-relevant uncertainty is better viewed as imprecision/credibility rather than fully observed frequencies, while the chance-constraint reformulation yields a practical deterministic equivalent at a target confidence level (Attar et al., 2025a; Keith & Ahner, 2021).

Consistent with common practice in agricultural quality/yield modeling, we assume α_1 and α_2 are normally distributed—an analytically convenient and empirically defensible approximation for biological attributes influenced by many small, quasi-independent factors. This assumption supports closed-form linearization of chance constraints and has precedent in crop-yield distribution studies that motivate (or critique) normality on biological and aggregation grounds; while actual crop yields can display negative skewness, arising in part from the concavity of biological production functions, Normal models remain a standard baseline and are frequently adopted for tractable optimization and risk analysis (Hennessy, 2009). Operationally, α_1 and α_2 enter the material-balance constraints for table-olive output (Eq. 3),

degraded-fruit allocation (Eq. 4), oil yields (Eqs. 5–7), and degraded-oil flows (Eq. 8); each affected constraint is transformed into a deterministic equivalent under the selected confidence level using the linearization detailed in Appendix C. Such linearization-based deterministic equivalents substantially improve tractability by converting inherently non-convex chance constraints into mixed-integer linear structures that can be solved efficiently with standard MILP solvers, while preserving provable approximation quality (Chen et al., 2024; Cui et al., 2025).

It is worth noting that a key methodological feature of the proposed chance-constraint linearization approach (detailed in Lemma C1, Appendix C) lies in its handling of Left-Hand Side (LHS) coefficient uncertainty. Typically, converting LHS uncertainty (where random parameters multiply decision variables) incorporates the variables' variances and covariances, often resulting in complex, non-linear second-order cone constraints. However, the proposed approach exploits the inherent structural specifications of the CLSC material balance equations. Because the uncertain quality parameter acts as a shared, multiplicative scalar across a sum of continuous variables, the standard deviation terms remain linear. This formulation naturally avoids second-order cone complexities, allowing the deterministic equivalent to be solved efficiently using standard MILP solvers without sacrificing mathematical rigor. This tractable transformation of LHS uncertainty represents a significant methodological contribution that can be readily adapted and applied to other CLSC networks facing similar yield or conversion uncertainties.

5 Solution approach

The proposed olive CLSC is formulated as a bi-objective MILP that minimizes total cost and water consumption. The integration of facility-location decisions, closed-loop material flows, water recovery mechanisms, and possibilistic uncertainty results in a large-scale and computationally challenging optimization problem. Although exact optimization methods can provide optimal solutions for small-scale instances, their computational performance would naturally deteriorate as network size increases. Therefore, a combination of exact and metaheuristic solution approaches is adopted in this study. Exact methods are first employed to validate the model and generate benchmark solutions for small instances, while several multi-objective metaheuristics are subsequently utilized to efficiently solve medium and large problems and approximate the Pareto frontier of cost–water trade-offs.

5.1 Test problems

This study considers twelve test problems in different sizes, among which problems 3, 5, and 12 are derived from a real-world case study of an Iranian olive orchard. The dimensions of the problem including gardens (G), processing plants (O), mills (M), packaging and distribution centers (D), recovery centers (R), fertilizer manufacturing centers (K), animal food industries (A), wastewater treatment plants (T), retailers (S), restaurants (H), food industries (F), cosmetic industries (C), pharmaceutical industries (E), products (P), packages (N), and water (W), are depicted in the Table 3. Other parameters and their values for initiating the problem are given in Table A2 (Appendix A). A Windows 10 laptop with a Core i5 2.7 GHz CPU and 12 GB RAM is used for all experiments reported in this study.

Case Study: Iran currently cultivates more than 90,000 hectares of olive orchards and reports an annual consumption of over 200,000 tons of green table olives and approximately

Table 3 Test problems settings

Test problem	<i>G</i>	<i>O</i>	<i>M</i>	<i>D</i>	<i>R</i>	<i>K</i>	<i>A</i>	<i>T</i>	<i>S</i>	<i>H</i>	<i>F</i>	<i>C</i>	<i>E</i>	<i>P</i>	<i>N</i>	<i>W</i>
1	8	2	2	2	1	1	1	1	1	1	2	2	2	8	14	4
2	16	4	4	4	2	2	2	2	2	2	4	4	4	8	14	4
3	25	6	7	6	5	4	6	2	4	3	6	6	6	8	14	4
4	30	8	8	6	6	6	6	4	4	4	8	8	8	8	14	4
5	34	12	12	10	10	8	8	5	10	10	10	10	10	8	14	4
6	50	20	20	18	15	10	10	6	20	20	12	12	12	8	14	4
7	70	30	30	22	25	12	12	7	30	30	15	15	15	8	14	4
8	90	40	40	35	35	14	14	8	40	40	18	18	18	8	14	4
9	110	50	50	45	45	16	16	9	60	60	22	22	22	8	14	4
10	130	60	60	50	55	20	20	10	80	80	25	25	25	8	14	4
11	150	70	70	55	65	22	22	11	100	100	30	30	30	8	14	4
Case study	250	50	60	65	80	24	24	12	500	500	120	100	100	8	14	4

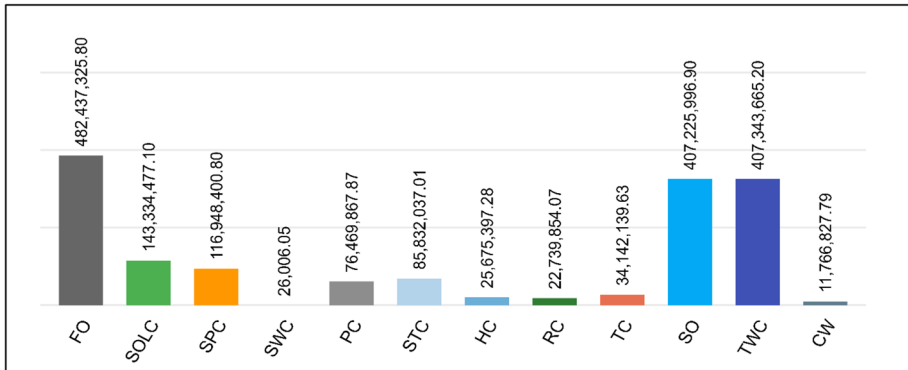


Fig. 3 Economic and water results of olive supply chain (FO: First Objective; SOLC: Supplying Olive Cost; SPC: Supplying Package Cost; SWC: Supplying Water Cost; PC: Process Cost; STC: Setup Cost; HC: Holding Cost; RC: Recovery Cost; TC: Transportation Cost; SO: Second Objective; TWC: Total Water Consumption; CW: Clarified Water)

12,000 tons of olive oil, with an estimated annual growth of about 5% (Iran Ministry of Agriculture, 2025). Olive production faces significant water stress for irrigation, olive processing, packaging, wastewater recovery, and harvesting situations, which require water-sustainable and process-efficient solutions. In this study, Iran is adopted as the case context to evaluate the behavior of the proposed supply chain model under different scenarios of process effectiveness, packaging quality, water collection and recovery processes, and harvest conditions to identify the leverage points and prioritized actions for mitigating water-related challenges in the olive value chain.

Figure 3 presents the economic and water performance of the modeled olive supply chain in Iran. The results clearly show that the procurement cost of olives and packages is the main cost driver, even though the purchasing prices assumed for both olives and packages are lower than international market levels. The cost of buying olives alone is about 167% higher than the total setup cost required to open 56 processing plants, 21 mills, 54 packaging and distribution centers, 41 recovery centers, 7 fertilizer manufacturing centers, and 7 animal-feed centers. This confirms that, from an investment perspective, the capital cost is not a barrier, and the payback period can be short, enabling investment in production and process efficiency.

A total of 248,358,764 units of different packages (small, medium, large, and bulk) is required to supply the national demand and enable inner transportation. Among them, 59.14% undergo recovery, which reflects a high defect rate and weak durability of packages. Small packaging sizes have the highest defect rates. For instance, bags used for extra virgin and virgin olive oil have a 97.5% defect rate because they cannot be directly reused and must go through recovery to re-enter the cycle, whereas tanks used for olive oil transport fail only about 3% of the time. This shows a trade-off between buying better packages with lower recovery frequency versus using cheaper packages that generate larger recovery flows, extra water consumption, and extra cost. Since the water used for package recovery accounts for only 16.1% of total process water and just 0.0096% of total water, including irrigation, the choice to use cheaper (but less durable) packages can still be economically reasonable, depending on whether decision-makers prioritize cost efficiency or water saving. Additionally, high defect-rate packages indicate misalignment between consumer preferences and the circular

principle (reuse cycle), meaning preference for small packages for convenience leads to inefficiency downstream and higher effort for material circularity. Holding and storage costs are also high, even though only about 2% of production is stored as excess. This is related to the biological fact that olives are fast-spoiling and require controlled storage, so the cost is driven by perishability rather than by volume.

Regarding water, well over 407 million USD/m³ of water is consumed to meet the annual demand of 200,000 tons of table olives and 13,000 tons of olive oil in Iran. Only about 0.02% of this usage can be recovered when irrigation is included, indicating that agricultural water dominates total water consumption. However, if irrigation is excluded and only processed water is considered, more than 48% of the used water can be recovered (after accounting for 15% collection loss). The result means that process-water recovery is not significant in terms of absolute volume reduction, but it is critical to avoid environmental damage, to reduce treatment penalties, and to enable industrial symbiosis using rich wastewater in the cosmetic and pharmaceutical industries. Hence, the figure reveals that the economic leverage lies mainly in procurement, the water leverage lies in agriculture, not in industry, and the circularity bottleneck lies in packaging design rather than in recovery capacity.

5.2 Exact method

To evaluate the problem on small instances, test problems P1–P3 are optimally solved utilizing the precise *LP-metric* aggregation method implemented in GAMS/CPLEX. The solution method uses the optimum value of each objective to reduce the distance of the final solution from the optimum space, as shown in Eq. (55). (see, for instance, Attar et al., 2025b for more details on the *LP-metrics* method).

$$\text{Min}Z = w_1 \times \left(\frac{f_1 - f_1^*}{f_1^*} \right) + w_2 \times \left(\frac{f_2 - f_2^*}{f_2^*} \right) \quad (55)$$

where, f_1^* and f_2^* are the ideal points of objective functions one and two, respectively, and w_1 and w_2 are the weights for the corresponding objectives. Table 4 presents the benchmark objective values corresponding to various w_1/w_2 ratios (i.e., fixed w_2 , varying w_1), which serve as a reference for evaluating solution quality.

Beyond confirming monotonicity, Table 4 reveals where additional cost emphasis delivers diminishing returns. Across Problems 1–3, increasing the cost weight w_1 from 0.1 to 0.9 reduces Z_1 by ~6–7% while increasing Z_2 by ~9–10%, confirming a systematic but asymmetric trade-off in which water consumption worsens slightly faster than cost improves. For example, in Problem 1, Z_1 falls from 3,011,720 to 2,807,040 (– 6.8%) as Z_2 rises from 12,514,368 to 13,687,590 (+ 9.4%); Problems 2–3 show the same monotone behavior. Stepwise changes also suggest a mild “knee” around $w_1 \in [0.4, 0.6]$: beyond this range the marginal reduction in Z_1 per 0.1 increase in w_1 becomes smaller, while the marginal increase in Z_2 stays roughly steady; i.e., further emphasis on cost yields diminishing returns and a steeper penalty in water. Practically, if decision-makers value water savings nearly on par with cost, weights near the mid-range appear to offer the most balanced compromises, whereas very high w_1 prioritizations buy relatively little extra cost reduction at a disproportionate water cost.

To formally assess the scalability boundary of the exact approach, the *LP-metric* model (Eq. 55) was, submitted to GAMS/CPLEX under a uniform three-hour resource limit across all problem instances. For medium-sized instances (P5–P8), GAMS did not achieve mathematical optimality within the allotted time; however, it returned incumbent solutions with an

Table 4 Results of the exact method under various weight ratios for the two objectives

Test problem	Objective value	Weight of first objective (w_1)								
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
1	Z ₁	3,011,720	2,986,135	2,960,550	2,934,965	2,909,380	2,883,795	2,858,210	2,832,625	2,807,040
	Z ₂	12,514,368	12,661,021	12,791,379	12,938,032	13,100,979	13,247,632	13,394,285	13,540,937	13,687,590
2	Z ₁	6,023,440	5,972,270	5,921,100	5,869,930	5,818,760	5,767,590	5,716,420	5,665,250	5,614,080
	Z ₂	25,028,160	25,321,459	25,582,169	25,875,468	26,201,355	26,494,654	26,787,953	27,081,251	27,374,550
3	Z ₁	9,411,110	9,331,161	9,251,213	9,171,264	9,091,315	9,011,366	8,931,418	8,851,469	8,771,520
	Z ₂	39,107,520	39,565,811	39,973,181	40,431,473	40,940,685	41,398,976	41,857,268	42,315,559	42,773,850

optimality gap below 5%, indicating that the branch-and-bound search tree retained residual tractability at this scale. The situation deteriorates prominently for larger, more realistic problem configurations: i.e., for P9–P12, the exact solver failed to identify any feasible integer solution within the three-hour limit. To illustrate the severity of this breakdown, the resource limit for P12 was extended to nine hours, at which point GAMS returned a feasible solution carrying a 4% optimality gap, confirming that the computational burden grows superlinearly with instance size and renders the traditional exact optimization practically infeasible for realistic olive supply chain configurations.

This behavior is particularly consistent with the inherent computational complexity of multi-echelon, multi-objective mixed-integer network design problems. The simultaneous treatment of binary facility-location decisions across multiple echelons (gardens, processing plants, mills, distribution centers, recovery facilities, wastewater treatment plants, and packaging nodes), combined with the possibilistic reformulation of quality-yield constraints, gives rise to a large-scale MILP whose search space grows exponentially with the number of nodes and commodities. Indeed, supply chain network design problems of this class are NP-hard in general, and branch-and-bound solvers are known to scale poorly beyond moderately sized instances even with state-of-the-art commercial software (Pishvaei et al., 2010; Soleimani & Kannan, 2015). Several analogous studies on agricultural and agri-food closed-loop supply chains report the same convergence breakdown: Liao et al. (2020) resorted to metaheuristics for their citrus CLSC precisely because exact methods became intractable at realistic network scales, a finding echoed by Rajabi-Kafshgar et al. (2023) in a pistachio CLSC context and by Seydanlou et al. (2022) in a closely related bi-objective olive network.

Beyond the empirical solver results, a structural comparison further reinforces this conclusion. The network configurations reported in the studies cited above as evidence of computational intractability (Liao et al., 2020; Pishvaei et al., 2010; Soleimani & Kannan, 2015) are, in each case, considerably simpler in topology than the present model: they typically involve four to six distinct facility types connected through a single primary commodity flow. As Fig. 2 illustrates, the proposed olive CLSC simultaneously integrates 12+ interdependent facility types, including gardens, processing plants, mills, packaging and distribution centers, recovery centers, fertilizer manufacturing centers, animal food industries, wastewater treatment plants, and multiple downstream demand clusters, across three structurally coupled flow systems (olive product flows, packaging circularity, and water recovery). This structural complexity directly translates into a substantially larger search space for branch-and-bound solvers, both in terms of the number of binary location variables and the density of coupling constraints across echelons, which provides further a priori justification for the computational intractability observed in our experiments.

5.2.1 Alternative methods

Given the computational intractability of traditional exact optimization methods for large-scale, real-world instances of the proposed multi-objective CLSC model, exploring alternative solution methodologies is imperative. As demonstrated in the preceding analysis, the exponential growth of the search space and the dense structural complexity of the network render standard branch-and-bound solvers ineffective for realistic configurations. While this study ultimately adopts a suite of multi-objective metaheuristic algorithms—a choice justified by their successful application in comparable supply chain literature, as briefly discussed in the previous section—it is necessary to evaluate other advanced mathematical techniques. Therefore, this section briefly discusses the applicability and limitations of two primary

alternative approaches, i.e., decomposition-based methods and uncertainty-specific heuristic methodologies.

(i) Decomposition-based exact methods: Benders Decomposition (Benders, 2005) is a classical partitioning strategy in this group of methods that separates complicating binary variables, in this case the facility-location decisions, from the remaining continuous flow variables, solving a master MILP and a continuous sub-problem iteratively to convergence. The classical Benders decomposition is not computationally appealing for the proposed multi-objective model due to its dependence on a decomposition into weakly coupled sub-problems (see Rahmaniani et al., 2017, and Hosseini & Turner, 2025, for a comprehensive review and recent advances in cut generation). The current formulation dictates facility-location decisions that concurrently regulate olive-product distribution, packaging logistics, and water-recovery strategies across various levels. This establishes dense linking constraints and robust cross-system interactions, rendering commodity-wise, classic decomposition ineffective and diminishing the efficacy of the resultant Benders cuts. (see, for instance, Alshamsi & Diabat, 2018, for comparable observations on large-scale supply-chain reverse logistics problems).

On the other hand, Lagrangian Relaxation (Fisher, 2004) offers a complementary decomposition strategy by dualizing the coupling constraints and solving smaller sub-problems independently. However, for general MILPs without special structure, the Lagrangian duality gap is not guaranteed to be tight, particularly when the problem involves a large number of binary variables across heterogeneous facility types (Guignard, 2003). In the bi-objective setting, separate Lagrangian multiplier updates would be required for each scalarized sub-problem, and convergence to the true Pareto front is not ensured. Furthermore, the Column Generation method and its integer-programming extension, Branch & Price (Barnhart et al., 1998), are most naturally applicable to problems admitting a set-partitioning or set-covering master formulation in which each column represents a complete feasible configuration such as a delivery route or a shift pattern. The present model is a multi-commodity, multi-echelon node-arc network design problem with heterogeneous facility types and cross-system flow coupling; it does not possess the structural regularity required to define a tractable column-enumeration scheme, making Column Generation nontrivial to implement effectively and unlikely to yield computational advantages without substantial reformulation (see Dantzig & Wolfe, 1960; Barnhart et al., 1998; Lübbecke & Desrosiers, 2005, for the class of problems where Column Generation and Branch & Price are most effective).

(ii) Uncertainty-specific approaches: For models under uncertainty, Progressive Hedging (Rockafellar & Wets, 1991; Watson & Woodruff, 2011) is an established scenario-decomposition algorithm for multi-stage stochastic programs. However, this method is primarily designed for scenario-based stochastic programs with explicit recourse structure (Rockafellar, 2018), which the present model does not possess. As detailed in Sect. 4.4 and Appendix C, uncertainty in olive yield quality is handled through a possibilistic programming framework that models imprecision as credibility rather than frequency-based probability (Keith & Ahner, 2021), and the resulting chance constraints are converted into deterministic linear equivalents prior to optimization. The formulation is therefore a single-stage, single-period MILP with no recourse decisions and no scenario tree to decompose. Applying Progressive Hedging would require abandoning the possibilistic framework entirely in favor of a multi-scenario stochastic reformulation, which would introduce distributional assumptions that possibilistic programming deliberately avoids, and would fundamentally alter the nature of the uncertainty representation adopted in this study.

Taken together, the empirical evidence from our experiments in the previous section and the corroborating literature reinforce that multi-objective metaheuristic algorithms are not

merely a convenient shortcut but a computationally more suitable tool for solving bi-objective CLSCs at the scales required by real-world agricultural operations.

5.3 Metaheuristic algorithms

The solution approach adopted in this study relies on well-established multi-objective metaheuristic algorithms, including Non-dominated Sorting Genetic Algorithm II (NSGA-II), Non-dominated Ranking Genetic Algorithm (NRGA), Multi-objective Variable Neighborhood Search (MOVNS), and Multi-objective Simulated Annealing (MOSA). These methods were selected to provide a balanced representation of the evolutionary, trajectory-based, and neighborhood search paradigms in multi-objective optimization. Such diversity allows for a comprehensive evaluation of algorithmic performance in terms of convergence, solution quality, and computational efficiency. Each algorithm is applied to a range of test instances of varying size and structural complexity, reflecting different configurations of supply chain elements (e.g., number of gardens, processing facilities, and recovery centers). The goal is to generate high-quality approximations of the Pareto frontier, thereby enabling decision-makers to explore trade-offs between total cost and water consumption. The comparative analysis further facilitates the identification of the most suitable algorithmic framework for solving large-scale, bi-objective CLSC design problems of the type considered in this study.

It is important to emphasize that the contribution of this study does not reside in the development of the metaheuristic algorithms themselves, as NSGA-II, NRGA, MOVNS, and MOSA are well-established approaches in the optimization literature. Rather, the methodological contribution lies in their adaptation to the specific characteristics of the proposed closed-loop supply chain network, including the problem-specific solution representation, encoding and decoding procedures, and the rigorous parameter tuning conducted using the Taguchi method, as described in Sect. 5.3.3. These problem-tailored implementation choices enable an effective exploration of the solution space and the generation of high-quality Pareto solutions for realistic network scales.

5.3.1 The proposed algorithms

(i) **Multi-objective Simulated Annealing (MOSA):** Simulated annealing is highly regarded and cited since it was first proposed by Kirkpatrick et al. (1983). The foundation of this algorithm is based on setting the initial temperature and observing the changes. This characteristic was highly utilized in previous studies to obtain an optimum solution. Nevertheless, SA acts differently from population-based algorithms (Akbarpour et al., 2020). As a result, here, some other characteristics, such as mutation, swap, and reversion, are used to help SA to achieve better solutions. The pseudocode of the applied MOSA is presented in Algorithm B1 (Appendix B), in which the initial solution x is the original solution and x' is the newly developed solution.

(ii) **Multi-Objective Variable Neighborhood Search (MOVNS):** The Variable Neighborhood Search (VNS) introduced by Mladenović and Hansen (1997) is a metaheuristic method employed for the heuristic resolution of optimization problems. The approach involves systematically altering the neighborhood structure throughout the search process. It transitions from its current position in the solution space to a new one if an improvement has taken place or particular acceptance criteria are met. Numerous successful implementations of the multi-objective VNS approach have been documented to date. The adaptation of the VNS framework to multi-objective optimization problems presents substantial innovations in the

conceptualization and enhancement of solutions. The VNS can effectively manage multiple objectives concurrently by redefining the solution as an approximate set of efficient points identified during the search process. Consequently, the VNS can be extended to maintain an approximation set of Pareto solutions, thereby behaving similarly to population-based methods in multi-objective settings. By considering the approximate Pareto front as the current solution, we can seamlessly adapt various VNS variants to multi-objective problems. Algorithm B2 (Appendix B) illustrates the pseudocode for our proposed MOVNS algorithm.

(iii) **Genetic Algorithms with Non-dominated Sorting and Ranking:** NSGA-II and NRGa are among the other renowned metaheuristic algorithms that have been used for optimization in recent years (Kord & Samouei, 2023). Both algorithms start by generating populations of possible solutions. Then these solutions are inserted into the model to verify its optimality and feasibility. These two algorithms then employ two operators, namely crowding and sorting, to opt for the best-generated solutions. NRGa employs roulette wheel selection, and NSGA-II uses binary tournament selection in its next step. The next step would be mutation, which helps prevent the algorithms from getting trapped in local optima. Algorithm B3 (Appendix B) shows the two pseudocodes in one picture.

5.3.2 Encoding and decoding strategy

Among the various methods and strategies to encode and decode a plan, the priority-based method of Enstroem et al. (2025) is considered. Based on this method, a string to show the number of open facilities in each center is taken into account. A random number between (0,1) is considered for each cell and then sorted for each section. To better show the steps of this method, a small instance is employed to shape the chromosome. This will also guarantee that all the required conditions are met with the constraints. The instance includes (2,2, 3,1, 1,1, 1,1, 1,2) centers to encode (gardens,SDCs,OFs,FIs,PIs, CoIs, MCs, AFIs, ChIs, FCs). The structure of the chromosome is shown in Fig. 4. The first row presents the product flow between centers. Here, with reference to the network in Fig. 2, five different sections can be considered (see Fig. 4). The second row demonstrates the corresponding centers which

Section 1				Section 2										
g		d		d		o + fm + pm + lm + mm + am + cm								
0.90	0.52	0.68	0.95	0.19	0.07	0.63	0.59	0.35	0.58	0.74	0.96	0.98	0.78	0.53
2	1	1	2	2	2	5	4	1	3	6	8	9	7	2

Section 3						Section 4													
o			fm+pm+lm+mm			o+fm+pm+lm+mm+am+cm+d									f				
0.31	0.12	0.75	0.74	0.75	0.39	0.57	0.50	0.80	0.36	0.71	0.01	0.29	0.54	0.96	0.41	0.57	0.26	0.58	0.08
2	1	3	3	4	1	2	6	10	4	9	1	3	7	11	5	8	2	2	1

Section 5									
f					g				
0.80					0.55				
2					1				

Fig. 4 The proposed strategy for encoding

product must flow. The next row is generated using random number drawn from the range $[0,1]$. The last row defines the priority based on the decoding plan.

5.3.3 Tuning of hyperparameters

We use the Taguchi (1986) method to set and evaluate the hyperparameters of the considered algorithms. This method has been used extensively in previous studies (Akbarpour et al., 2020; Wang & Zhang, 2024). Using algorithm settings tuned to the problem ensures the comparison of the algorithms is fair (as default parameters may not be best suited for the problem at hand) and that each of the algorithms performs close to optimally. These settings contribute to the two factors to control noise and factor with the set of orthogonal arrays. The rule “smaller-is-better” is applied to choose the best levels of the algorithm with the objective function of minimization. The utilized level of the problem is presented in Table 5, where X is calculated as $X = g + d + o + fm + pm + lm + mm + am + cm + f$. Using Minitab application to analyze the Taguchi test problems enables extracting efficient experiments to find the optimum levels for each algorithm parameter faster. Here, thirty experiments are analyzed to generate the optimum responses, and the designed experiment for each algorithm is as follows:

$$L_9 : \text{NSGA} - \text{II}, \text{NRGA}, \text{MOSA} \quad L_{27} : \text{MOVNS}$$

Table 5 Corresponding levels of the algorithms

Algorithm	Factor	Factorslevels			Best level
		First level	Second level	Third level	
NSGA – II	A : Pc	0.75	0.85	0.95	0.75
	B: Pm	0.06	0.11	0.16	0.06
	C: N-pop	60	120	180	180
	D: Max-iteration	2X	3X	4X	4X
NRGA	A: Pc	0.65	0.75	0.85	0.75
	B: Pm	0.04	0.12	0.18	0.04
	C: N-pop	50	100	150	150
	D: Max-iteration	2X	3X	4X	4X
MOVNS	A: K_max (K)	5	6	7	6
	B: Max_Iteration (I)	150	250	350	250
	C: Selection_Strategy (S)	Crowding-distance	Random	Roulette-based	Roulette-based
	D: Shaking Strength (SS)	Reversion	Swap	Insertion	Reversion
MOSA	A: T ₀	35	45	55	55
	B: α	0.95	0.90	0.85	0.95
	C: Max-iteration	2X	3X	4X	3X

The signal to noise (S/N) ratio helps to identify the best levels in each method. The standard Eq. (56) for the signal to noise ratio is present as follows:

$$S/N = -10 \log \left(\sum (Y^2) / n \right) \quad (56)$$

where, Y is the response time and n the number of orthogonal arrays. Using Eq. (56), the best algorithm parameter settings are shown in Table 5.

5.3.4 Algorithm evaluation

The proposed metaheuristics are evaluated using two performance metrics, including MeanIdealDistance (MID) and CPU time. MID measures the average normalized distance of the acquired solutions from the ideal point in the objective space, thereby indicating the convergence and quality of the generated solutions, while the CPU time indicates computational efficiency. The test problems are categorized into small (1–4), medium (5–8), and large (9–12) instances, with the corresponding MID and CPU time values presented in Table 6. For the MID metric, we utilize Eq. (57), which has proved to be efficient for multi-objective programming (Goodarzian et al., 2024).

$$MID = \sum_{i=1}^n \sqrt{\left(\frac{f_i^1 - f_{best}^1}{f_{total}^{1max} - f_{total}^{1min}} \right)^2 + \left(\frac{f_i^2 - f_{best}^2}{f_{total}^{2max} - f_{total}^{2min}} \right)^2} / n \quad (57)$$

The MID and CPU times with lower values yield the best results, as shown in Table 6. The analysis of variance (ANOVA) is used here to compare these two metrics, and the corresponding plots are shown in Fig. 5. For these interval plots of each metaheuristic, 12 points have been obtained.

The results of MOSA in Fig. 5a and b are clearly superior in terms of CPU time, and NRGa is considered to be the best algorithm in terms of MID . Figure 6 depicts the sets of non-dominated (Pareto) solutions generated by the metaheuristics for three representative instances (Problems 3, 5, and 10), emphasizing the intrinsic trade-off between total cost and water consumption. Across all panels, the expected monotonic trade-off between total cost and total water consumption is preserved; however, the shape, dispersion, and algorithmic distribution of solutions vary markedly with instance size. Overall, the plots illustrate that algorithmic behavior becomes more distinguishable as problem complexity grows.

In the small instance (Fig. 6a), the Pareto front displays a compact, moderately sloped cloud of points, indicating only a mild trade-off between cost and water. Here, we can observe that: (i) NRGa (blue triangles) consistently identifies solutions with the lowest water consumption among all algorithms, regardless of total cost. Its cluster sits noticeably below the others, demonstrating superior water-efficiency; (ii) MOSA (red squares) and NSGA-II (yellow circles) occupy the central mass of the front, with similar cost ranges but slightly higher water consumption; And, (iii) MOVNS (green diamonds) exhibits the widest spread, but its solutions tend to be somewhat higher in water use and do not dominate NRGa or MOSA. Furthermore, the shape of the front suggests a gentle knee, meaning a moderate increase in cost yields diminishing gains in water reduction. In small networks, therefore, the trade-off is relatively forgiving, enabling decision-makers to achieve balanced improvements without severe cost penalties.

For the medium instance (Fig. 6b), the Pareto front becomes more pronounced and more steeply rising, indicating stronger interaction between cost and water objectives. Here, NRGa again achieves the lowest water levels, forming the left-lower envelope of the front. NRGa's

Table 6 Compare results of each algorithm (MID: mean ideal distance, CPU time: computational time, seconds)

Test problem	MID		CPU time					
	NSGA – II	NRGA	MOVNS	MOSA	NSGA – II	NRGA	MOVNS	MOSA
1	1.50	2.38	1.42	3.94	63.45	70.13	108.17	18.65
2	1.13	1.18	1.18	2.34	138.79	155.33	260.44	29.32
3	2.13	2.04	2.95	2.15	239.77	265.01	488.16	70.10
4	3.64	2.13	2.34	5.55	363.24	411.55	751.84	120.65
5	3.73	2.63	3.72	3.24	895.53	1002.78	2032.69	224.81
6	3.21	2.83	3.36	5.08	1187.69	1285.87	2266.98	460.88
7	5.06	5.49	4.28	4.86	1385.09	1547.97	2654.92	510.57
8	5.93	5.71	5.39	5.02	2327.55	2615.35	4635.05	810.52
9	4.89	4.12	6.78	5.19	5573.30	5058.55	7310.12	1080.43
10	4.00	3.74	10.39	5.90	6137.65	5645.28	9232.27	1170.88
11	5.88	4.09	4.96	8.30	10,535.63	10,310.78	13,181.41	1280.74
12	4.91	6.45	5.64	8.10	13,447.58	14,364.68	18,150.34	1650.02

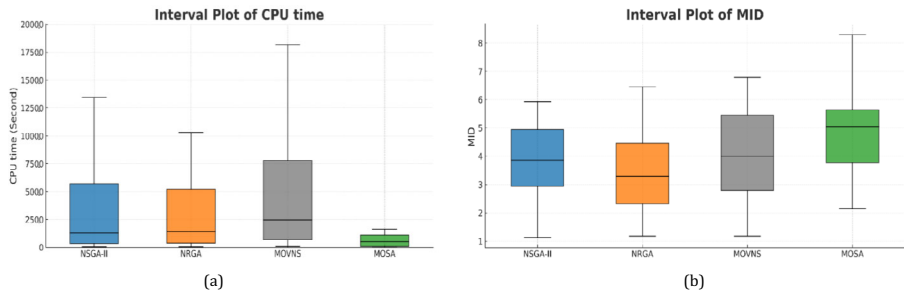


Fig. 5 **a** The interval plot of CPU time (at 95% confidence level), **b** the interval plot of mean ideal distance (MID) (at 95% confidence level)

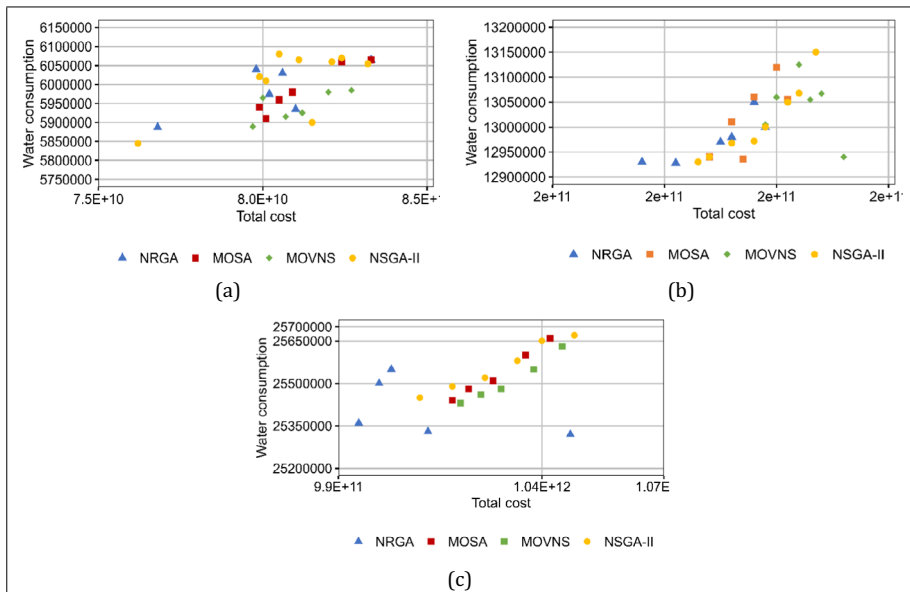


Fig. 6 Pareto solution for **a** test problem 3; **b** test problem 5; **c** test problem 10

solutions are consistently superior in water consumption across almost the entire cost range. MOSA and NSGA-II now show clear upward slopes, reflecting a stronger cost–water coupling: as cost decreases, water consumption rises more sharply relative to Fig. 6a. On the other hand, MOVNS solutions appear higher in water consumption and occasionally cluster near the upper boundary of the front. MOVNS maintains diversity but provides less competitive trade-offs compared to NRGA or MOSA. Overall, this behavior reflects rising structural complexity as more facilities and flows are included.

In the large instance (Fig. 6c), the trade-off becomes most severe: the Pareto cloud steepens considerably, showing that even minor water reduction demands disproportionately large cost increases. This figure shows that: (i) NRGA again produces the lowest water-consuming solutions, clearly forming the dominant lower edge of the Pareto frontier—more distinctly than in (a) and (b); (ii) MOSA and MOVNS both cluster at higher cost–water combinations,

following a roughly linear upward trend, with MOVNS occasionally reaching the highest water consumption values; and (iii) NSGA-II again populates the upper region, showing higher water usage than NRGa or MOSA for similar cost levels. The slope of the front in (c) is the steepest among the three subfigures, reflecting a network in which efficiency improvements become increasingly costly. The knee point is located close to the NRGa cluster, indicating that for large systems, the most attractive solutions (balancing cost and water) lie almost exclusively among NRGa's outputs.

Across all three instances of Fig. 6, a consistent pattern emerges: (1) NRGa produces clearly superior (lower-water) solutions, especially as the instance size grows; (2) Trade-offs become sharper with scale, i.e., from a gentle slope in (a) to a steep, near-linear frontier in (c); and (3) Algorithmic differentiation increases with problem size: in small problems most algorithms cluster tightly; in large instances their performance spreads widely. From a practical and Managerial standpoint, these patterns imply that:

- For small systems, multiple algorithms provide workable solutions with modest trade-offs.
- For larger, real-world systems, only a subset of methods (particularly NRGa) can reliably navigate the sharper Pareto structure to find high-quality solutions.
- Decision-makers should expect dramatically rising marginal costs when pushing for aggressive water savings in large agricultural CLSCs.

5.4 Impact of chance constraints on algorithmic performance

A key contribution of our framework is the use of chance constraints to capture yield-quality uncertainty via the stochastic quality rates of early- and main-harvest olives, α_1 and α_2 (normal distributions with mean μ_1 and standard deviation σ_1). These parameters enter the material-balance constraints for table-olive and oil-output streams and propagate costs and water use through procurement, processing, transport, recovery, and wastewater subsystems. While Sect. 5.3 earlier compared metaheuristics on MID and CPU time, we now quantify how the chance-constraint modeling itself changes solution quality and algorithm behavior under varying uncertainty and reliability requirements. We evaluate two questions on benchmark instances P1–P3 (small to medium sizes): (1) Uncertainty sensitivity: How do outcomes change as the *volatility* of α_1 , α_2 increases? (2) Algorithmic resilience: Do solver rankings (NRGa, NSGA-II, MOVNS, MOSA) change when uncertainty and reliability tighten the feasible set?

The variation factors considered in these experiments are twofold:

- Confidence level (ϵ'):** We define $\epsilon' = 1 - \epsilon$, with $\epsilon' \in \{0.80, 0.95, 0.99\}$. Higher values of ϵ' tighten the deterministic equivalents derived in Appendix C and consequently reduce the feasible region. This tightening applies to all constraints involving the stochastic quality parameters
- Uncertainty scale:** We scale the standard deviation σ_1 by $\gamma \in \{0.5, 1.0, 1.5\}$ for both α_1 , α_2 . Such scaling scenarios explore more/less volatile harvest quality.

For each instance and factor combination, the four metaheuristics (NSGA-II, NRGa, MOVNS, and MOSA) were executed over 10 independent seeds. We report three key performance metrics: (i) the Mean Ideal Distance (MID) as a proxy for the convergence and quality of the obtained Pareto front; (ii) the CPU time required for convergence, indicating computational efficiency; and (iii) the ex-post violation rate, which measures the empirical reliability of the solutions when validated against 500 Monte Carlo draws of the stochastic

Table 7 Aggregated Medians (IQRs) of MID, CPU Time, and Violation Rates Across P1–P3 Under Varying ϵ' and γ

ϵ'	γ	Metric	NSGA-II	NRGA	MOVNS	MOSA
0.80	0.5	MID	1.9 (1.5–2.5)	1.7 (1.4–2.2)	2.2 (1.8–2.9)	3.4 (2.8–4.3)
		CPU (s)	140 (100–190)	150 (110–210)	260 (210–340)	30 (20–45)
		Violation (%)	4 (2–7)	3 (1–6)	6 (3–9)	9 (5–14)
	1.0	MID	2.5 (2.0–3.2)	2.2 (1.8–2.8)	2.8 (2.3–3.6)	4.2 (3.5–5.2)
		CPU (s)	190 (140–260)	200 (150–280)	350 (280–450)	40 (25–60)
		Violation (%)	6 (3–10)	5 (2–9)	8 (4–12)	12 (7–18)
	1.5	MID	3.3 (2.6–4.2)	2.9 (2.3–3.6)	3.9 (3.1–5.0)	5.9 (4.7–7.3)
		CPU (s)	265 (200–360)	280 (220–380)	490 (390–630)	55 (35–85)
		Violation (%)	9 (6–14)	8 (5–13)	12 (8–18)	18 (12–25)
0.95	0.5	MID	2.3 (1.8–2.9)	2.4 (1.9–3.0)	3.0 (2.4–3.8)	4.0 (3.2–5.0)
		CPU (s)	205 (160–280)	220 (170–300)	380 (300–490)	45 (30–65)
		Violation (%)	8 (5–13)	9 (5–14)	12 (8–18)	16 (10–23)
	1.0	MID	2.9 (2.3–3.7)	3.0 (2.4–3.8)	3.8 (3.0–4.8)	5.0 (4.0–6.2)
		CPU (s)	260 (200–350)	280 (220–380)	480 (380–620)	55 (35–85)
		Violation (%)	11 (6–17)	12 (7–18)	16 (10–23)	20 (13–28)
	1.5	MID	4.1 (3.3–5.1)	4.2 (3.4–5.3)	5.3 (4.2–6.7)	7.0 (5.6–8.7)
		CPU (s)	365 (280–490)	390 (310–530)	670 (530–870)	75 (50–110)
		Violation (%)	15 (10–22)	17 (11–24)	22 (15–31)	28 (19–38)
0.99	0.5	MID	2.8 (2.2–3.5)	3.0 (2.4–3.8)	4.1 (3.3–5.2)	5.2 (4.2–6.5)
		CPU (s)	280 (220–370)	300 (240–400)	510 (410–660)	60 (40–90)
		Violation (%)	13 (8–19)	14 (9–20)	19 (13–27)	24 (16–33)
	1.0	MID	3.5 (2.8–4.5)	3.8 (3.0–4.8)	5.2 (4.2–6.5)	6.5 (5.2–8.0)

Table 7 (continued)

ϵ'	γ	Metric	NSGA-II	NRGA	MOVNS	MOSA
1.5		CPU (s)	350 (280–460)	380 (300–500)	650 (520–850)	75 (50–110)
		Violation (%)	16 (10–23)	18 (12–25)	24 (17–32)	30 (22–40)
		MID	4.9 (3.9–6.1)	5.3 (4.2–6.6)	7.3 (5.8–9.1)	9.1 (7.3–11.2)
		CPU (s)	490 (390–640)	530 (420–690)	910 (730–1180)	105 (75–150)
		Violation (%)	22 (16–31)	25 (18–35)	34 (24–46)	42 (30–55)

parameters. Table 7 aggregates the medians and interquartile ranges (IQRs) of these metrics across instances P1–P3, illustrating the performance under varying confidence levels (ϵ') at a varying uncertainty scale (γ).

The results presented in Table 7 comprehensively validate our initial hypotheses, providing clear answers to our three primary research questions regarding uncertainty sensitivity, algorithmic resilience, and feasibility reliability. The results reveal a strong synergistic effect between the confidence level (ϵ') and the uncertainty scale (γ), where tightening either factor degrades performance, but tightening both simultaneously leads to a disproportionate increase in problem difficulty.

First, the impact of uncertainty sensitivity is pronounced across all algorithms. As the problem becomes more constrained (moving from the most lenient setting, i.e., $\epsilon' = 0.80$, $\gamma = 0.5$, to the most restrictive, i.e., $\epsilon' = 0.99$, $\gamma = 1.5$), the reported performance deteriorates significantly. For instance, the median MID for the best-performing algorithm, NRGa, degrades by over 210% (from 1.7 to 5.3), while its median CPU time increases by 250% (from 150 to 530s). This trend, also observed across all metaheuristics, reflects the severe shrinkage and fragmentation of the feasible search space, which heightens the computational burden of identifying high-quality, robust solutions.

Second, the results reveal critical shifts in algorithmic resilience and relative performance rankings. While NRGa and NSGA-II consistently deliver the highest-quality solutions, their dominance becomes more nuanced under pressure. At lower to moderate constraint levels, NRGa often maintains a slight edge in solution quality (MID). However, at the highest confidence level of $\epsilon' = 0.99$, NSGA-II consistently achieves a lower median MID than NRGa across all uncertainty scales (e.g., 3.5 vs. 3.8 at $\gamma = 1.0$). This reversal suggests that NSGA-II's binary tournament selection mechanism offers greater resilience against premature convergence when the feasible region becomes exceptionally restrictive. In stark contrast, the performance gap between the genetic algorithms and the other methods widens considerably. MOVNS, a local search-based method, experiences a disproportionate increase in both MID and CPU time, as its neighborhood search falters in disconnected feasible pockets. MOSA, despite remaining the fastest algorithm, suffers a severe drop in solution quality, with its median MID increasing by over 160% in the most extreme scenario. The widening Interquartile Ranges (IQRs) for both MOSA and MOVNS further indicate a sharp decline in their performance consistency as problem difficulty increases.

Third, the analysis of feasibility reliability yields a crucial insight into the practical limits of the chance-constrained model. The ex-post violation rates test whether the solutions empirically satisfy the model's probabilistic guarantees. At the $\epsilon' = 0.80$ confidence level (implying a target violation rate ≤ 20), all algorithms produce solutions whose median violation rates (3–12% at $\gamma = 1.0$) are comfortably below the threshold. However, for $\epsilon' = 0.95$ (target ≤ 5) and $\epsilon' = 0.99$ (target ≤ 1), all four metaheuristics return solutions that consistently violate the chance constraints at rates far exceeding the target levels. This finding highlights a potential gap between the theoretical guarantees of the deterministic equivalent formulation and its empirical performance under ex-post simulation, especially at very high confidence levels. It suggests that while the model effectively steers the search towards more robust solutions, achieving near-certain feasibility may require more advanced constraint-handling techniques or more conservative formulations.

In summary, these results affirm the viability of using metaheuristics for this class of uncertainty-aware supply chain problems. For practitioners, the choice of algorithm involves a clear trade-off: NRGa and NSGA-II are superior for generating high-quality, reliable solutions, with NSGA-II showing particular strength in highly constrained, high-uncertainty

settings. Conversely, MOSA is best suited for time-critical applications where approximate solution quality is acceptable, and the constraints are less severe.

6 Additional analysis and implications

To further assess the proposed bi-objective model and derive actionable managerial insights, this section presents a comprehensive sensitivity analysis. Given the inherent uncertainties and operational complexities within the olive supply chain, it is crucial to evaluate how fluctuations in key parameters impact the network's overall economic and environmental performance. Specifically, we investigate the system's responsiveness to changes in three critical operational levers: degraded harvested olive levels, the water collection loss portion, and package quality. By systematically analyzing these leverage points, this section illuminates the underlying trade-offs between cost efficiency and water circularity, ultimately providing strategic guidance for decision-makers navigating real-world supply chain disruptions.

6.1 Sensitivity analysis on degraded harvested olive level

The quality of harvested olive fruits (green, purple, and black olives) can be affected by weather conditions, water tension, pests, and harvesting mechanisms that can generate more degraded olive fruits. As mentioned, degraded green olives cannot be used for green table olive production but can be used for olive oil production with about 15% oiling percentage, while degraded black and purple olives cannot be used for the oiling process and can be directed for fertilizer or animal food production (Spada et al., 2025). In this case, we changed the degraded level of harvested green and black olives from 8 to 22%.

A 10% rise in degradation results in more than a 3% increase in total cost (Fig. 7), driven by larger procurement volume, more washing, more packaging use, more recovery effort, and more transport and storage, even though Iran is not a major olive consumer worldwide (International Olive Council, 2025). This means degradation is not only a quality loss at the orchard level but also a significant inefficiency downstream. What is more concerning is that olives represent a direct and irreversible water loss at the cultivation level, since irrigation water cannot be recovered, and industrial water recovery does not compensate for poor cultivation performance. Producing twice the degraded olives at 22% degradation sharply increases total water consumption while offering no additional value. Therefore, investment in cultivation quality (irrigation scheduling, pest management, soil improvement, and gentle harvesting) is a more powerful solution than industrial optimization or recovery technologies.

Improving harvested quality simultaneously reduces total cost, lowers water consumption, prevents unnecessary packaging and waste recovery, and reduces environmental stress. Additionally, since irrigation water is not recoverable, every kilogram of degraded fruit represents a permanent water loss. Therefore, public incentives, supporting agriculture practices, and fiscal instruments should target on-farm improvements in the case study (irrigation management, pest monitoring, and harvest mechanization) as the primary policy instrument to reduce national water demand and stabilize the olive supply chain. However, industrial water recovery remains useful to control water pollution and improve process efficiency, and additionally provides rich wastewater economic and environmental benefits, not as a solution for low-quality cultivation.

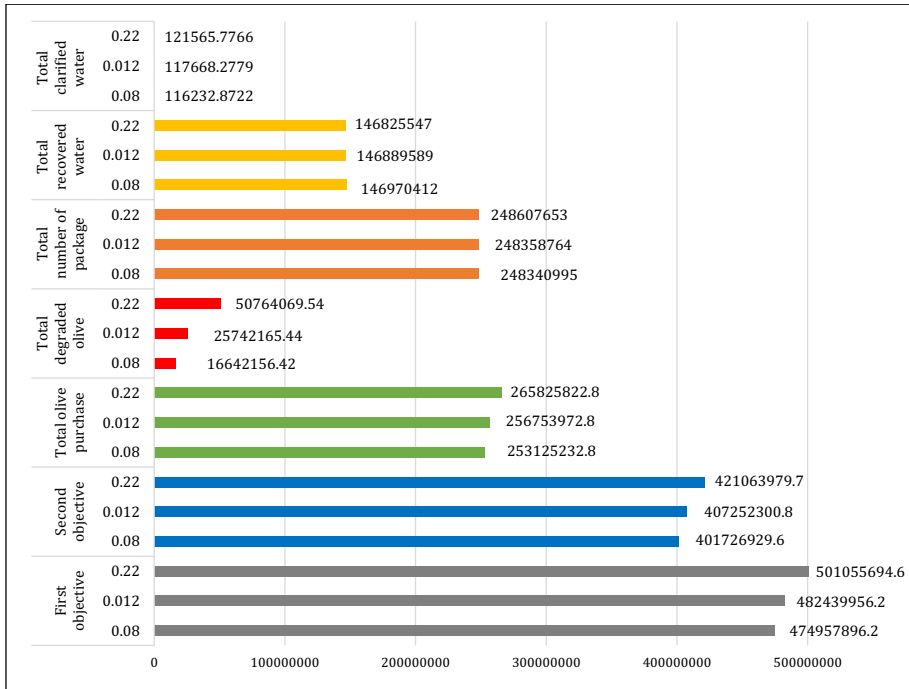


Fig. 7 Degraded level impact on olive supply chain performance

In short, the cheapest and most effective water-saving and cost-saving measure in the olive supply chain is not recovering more water inside factories but preventing degradation at the farm. Therefore, managers should prioritize reducing degradation at the orchard stage rather than processing facility performance enhancement through direct actions such as regulated irrigation, pest control, and gentler harvesting, leading to not only higher economic and environmental benefits but also faster payback.

6.2 Sensitivity analysis on water collection loss portion

As shown in Table 8, even though the earlier analysis showed that irrigation control and orchard quality are far more decisive for water consumption and cost, industrial water recovery still works as a lever to reduce land and water pollution from process wastewater. Changing the water collection loss rate proves that its direct numerical effect on total cost and total water is negligible. This does not imply that the decision is unimportant; it only means that the model does not internalize the environmental damage. In reality, a higher loss rate means more untreated acidic and salty effluent entering soil and groundwater, which converts into future penalties, liability, and regulatory pressure. Another important hidden effect is that a high collection loss directly reduces the potential for circular water use. When less water is collected and treated, there is less available to replace fresh tap water or groundwater in the next cycle, particularly in regions facing rising water prices or rationing. Moreover, excluding irrigation water, a higher loss forces the withdrawal of more freshwater for industrial needs and increases tension on processing facilities in already water-stressed conditions.

Table 8 Water collection loss analysis on olive supply chain performance

Variables	Water collection loss portion	Value
First objective	5%	0.0026% ↑
	15%	482, 439, 956.2
	20%	0.0051% ↓
	30%	0.0103% ↓
Second objective	5%	0.0016% ↓
	15%	407, 252, 300.8
	20%	0.0032% ↑
	30%	0.0064% ↑
Water collection volume	5%	5.5556% ↑
	15%	117, 668.2779
	20%	11.11% ↓
	30%	22.22% ↓
Supplying water cost	5%	2.28% ↓
	15%	28, 636.44575
	20%	4.57% ↑
	30%	9.13% ↑

That is, while improving on-farm water efficiency is the dominant solution for reducing absolute water use, improving water collection inside facilities is still justified as a pollution-mitigation, circular-water-preservation, and risk-prevention measure. It does not bring large numerical savings today, but it protects future compliance, reduces freshwater dependence, and maintains circularity in processing operations. Therefore, decision makers in processing facilities should not wait until regulations tighten but should proactively invest in water collection systems to reduce dependency on fresh water and avoid future compliance shocks. In Iran, where access to industrial water is already limited and seasonal rationing is becoming more common, improving collection and reuse is not only good for the environment, but it is also a smart way to protect against operational problems and rising costs in the future.

6.3 Sensitivity analysis on package quality

As expected, lower package quality causes a larger share of packages to enter the recovery cycle, which increases recovery cost, recovery water use, and the need for handling and treatment to avoid environmental damage. This means the burden is not only in the purchasing cost line but also in hidden downstream costs: more water must be collected and treated, and reverse logistics must process more defective units, which increases congestion and instability in the loop. When package quality is improved (lower defective rate), reuse increases, but the immediate model output shows that total cost (7.78%) rises due to the higher purchasing cost of durable packages, while the reduction in total water consumption appears minimal (0.001%), as shown in Table 9.

However, this small water reduction reflects only total system water, including irrigation. In reality, higher-quality packages reduce process-water consumption for recovery and reduce the need for reverse-flow handling and wastewater treatment, which are hidden operational

Table 9 Package quality analysis on olive supply chain performance

Variables	Defective rate	Value
First objective	10% ↓	7.78% ↑
	Normal	482,439,956.2
	10% ↑	2.80% ↑
Second objective	10% ↓	0.001% ↓
	Normal	407,252,300.8
	10% ↑	0.0002% ↑
Total package purchase	10% ↓	5.48% ↑
	Normal	248,358,764
	10% ↑	6.30% ↓
Total recovered package	10% ↓	8.11% ↓
	Normal	146,889,589
	10% ↑	6.26% ↑
Total recovery water	10% ↓	9.35% ↓
	Normal	39,005.20788
	10% ↑	24.74% ↑
Package purchasing cost	10% ↓	18.65% ↑
	Normal	116,948,400.8
	10% ↑	8.65% ↓
Total recovery cost	10% ↓	15.79% ↓
	Normal	22,739,854.07
	10% ↑	23.07% ↑

and environmental costs not fully captured in the economic objective. Although the economic evaluation shows that improving package quality increases procurement cost, it simultaneously creates the opportunity to mitigate environmental damage, reduce heavy recovery water consumption, and reduce the recovery workload. Higher-quality packages lower the pressure on collection infrastructure, decrease the frequency of reprocessing, and strengthen the overall movement toward circularity and sustainable material use. In this sense, the result does not reject high-quality packaging; instead, it highlights its value as a strategic choice for long-term stability and sustainability rather than a short-term cost decision. On the other hand, low-quality packages (higher defective rate) should not be considered as a cost-saving choice when they cause a 2.8% rise in total cost and a 0.0002% rise in water consumption through increasing recovery efforts to confront low-quality package waste. Therefore, investment in higher-quality packages (longer-life packages that can be reused) not only reduces reverse logistics pressure but also prevents environmental impacts such as virgin resource consumption, energy and water consumption for handling package waste recovery, and systematic costs that do not appear in package purchasing costs but emerge in recovery processes.

This all shows that package quality can be a lever to control cost and environmental impact in the long term by increasing their life through reusing high-quality packages. In other words, improving package durability is a practical tool to reduce environmental leakage and reverse-logistics pressure without interfering with production activities. Higher quality reduces recovery water use and treatment demand, which matters in countries like Iran, where industrial water stress and regulatory pressure on discharge are increasing. That can

be offered through introducing minimum durability, incentives for reusable packaging, and industrial water consumption pricing.

6.4 Managerial implications

Considering the validation results synthesized in the previous subsections, as well as the algorithmic performance and behavioral insights derived from the proposed olive supply chain model, what can obviously be found is the effect of olive's characteristics on the proposed supply chain network. The unique characteristic of the olive tree makes these products different from others in the agriculture industry. Unlike conventional agri-food supply chains, the olive network is shaped by distinct product flows, customer clusters, and reverse logistics requirements, all of which are influenced by the biological traits of olives and their sensitivity to environmental conditions. The model reveals that procurement and operational costs (particularly those associated with olive cultivation and processing) are the dominant cost drivers, surpassing setup and transportation expenses. Moreover, climate variability and pest pressure significantly affect crop quality and yield, amplifying both cost and water consumption across the supply chain. While the olive tree is known for its resilience, the degradation of fruit due to suboptimal growing conditions or harvesting practices leads to increased waste, reduced product quality, and higher resource consumption.

From a managerial perspective, the results suggest that reducing degradation at the orchard level through improved irrigation scheduling, pest control, and harvesting techniques, offers the most effective leverage for minimizing both cost and water use. Investments in cultivation quality yield system-wide benefits, including reduced packaging demand, lower recovery costs, and decreased environmental impact. In contrast to many supply chain studies that emphasize downstream optimization, this research highlights the upstream stages (particularly cultivation) as the most critical intervention points. The model also demonstrates that packaging quality plays a strategic role in circularity and cost control. While high-quality packaging incurs greater upfront costs, it reduces reverse logistics pressure, water use, and environmental risks associated with waste handling. Furthermore, the analysis of water collection loss rates reveals that although their direct numerical impact on total cost and water consumption is limited, they have significant implications for environmental compliance and long-term sustainability. Poor water collection efficiency increases untreated discharge, reduces the potential for reuse, and heightens regulatory risk—especially in water-stressed regions like Iran. Additionally, storage in the proposed model represents a capacity buffer and surplus-balancing term, aligning with strategic planning rather than time-based inventory management. This structure supports not only long-term network configuration but also strategic facility and capacity decisions, while operational harvest timing and time-dependent inventory management are left for future tactical decisions that can be considered as an extension to this approach.

The model also treats the rich wastewater streams directed to cosmetic and pharmaceutical uses as managed, revenue-generating outputs rather than disposal burdens. Within the framework of circular bio-economy business models, this routing represents a form of cascading, whereby a single waste stream is directed toward progressively higher-value applications instead of a single terminal use (Donner & de Vries, 2021). Interpreting wastewater valorization through the lens of cascading provides managers with a concrete strategic mechanism. It links operational decisions regarding the allocation of recovered water and by-products across the network to broader circular bio-economy objectives, while clarifying how value can be

captured beyond conventional cost reductions and water-saving benefits. Overall, the proposed model provides a comprehensive framework for evaluating trade-offs and identifying leverage points in a complex, water-sensitive agricultural supply chain. It enables decision-makers to move beyond short-term cost minimization and adopt a stability-oriented approach that integrates environmental stewardship, resource efficiency, and long-term resilience.

7 Conclusions

This study proposed and evaluated a bi-objective, closed-loop supply chain for the olive sector that jointly optimizes economic and water use aspects while explicitly integrating forward and reverse flows, packaging circulation, and both wastewater and rich-water streams. By embedding realistic yield-related uncertainty via possibilistic programming and solving large instances with tuned multi-objective metaheuristics, we demonstrated that a practically sized, water-aware CLSC can be optimized at scale and applied to real contexts. The case study based on the Iranian olive network validates the model's ability to identify dominant leverage points and trade-offs that are often overlooked in conventional forward-only or single-objective formulations.

Here, several key insights emerged: (1) the procurement costs (particularly for olives and packaging) are the primary cost drivers, far exceeding capital expenditure for facility setup. This suggests that investment in process upgrades and quality systems is economically feasible and not a major barrier; (2) the irrigation accounts for over 99% of total water consumption, indicating that industrial water-saving measures, while beneficial for compliance and risk mitigation, have limited impact on the overall water footprint. Recovery of processing water (~ 48%) serves more as a regulatory and circularity tool than a volumetric solution; (3) the high degraded-fruit shares (from pests, climate, or handling) amplify costs and water use across procurement, washing, storage, packaging, and transport; (4) likewise, high package-defect rates inflate reverse flows, recovery workload, and recovery water, exposing hidden system costs of low-durability packaging; and finally, (5) weak water collection systems (losses) erode circularity and elevate untreated discharge risk even when its effect on total-water accounting appears numerically small. These insights and patterns are consistent with and illuminated by the model's integrated representation of product, package, and water networks.

From a managerial standpoint, the most effective lever for water saving lies upstream in cultivation (through improved irrigation design, pest control, soil health, and harvesting practices) rather than inside processing plants. Packaging upgrades should be viewed not merely as immediate cost-savers but as strategic investments in long-term stability and sustainability that would reduce reverse burden, failure rates, and environmental pressure. In turn, better packaging and industrial water recovery reinforce each other by enabling safer, more efficient reuse streams. Internal investment in water collection and treatment is justified as a hedging and license-to-operate measure: it curbs land and regulatory risks despite limited short-run impact on total-water metrics at current prices. Crucially, improving orchard quality lowers system stress end-to-end, simultaneously reducing cost, water use, recovery needs, and environmental impact. These findings advocate for separating "short-term cost optics" from "long-term stability": actions that may appear neutral on the short-term cost (e.g., improving quality, collection, or recovery) often deliver substantial value through risk reduction and enhanced system robustness.

From the theoretical aspects, the work advances olive supply-chain modeling by unifying forward, reverse, packaging, and water subsystems in a single dual-objective model with uncertainty, an integration rarely treated conjointly in prior studies. It also shows that multi-objective metaheuristics, after systematic parameter tuning, can deliver competitive Pareto fronts for large, water-aware CLSCs and can be cross-checked on small instances with exact methods. From a methodological perspective, although advanced exact and decomposition-based methods such as Benders decomposition, column generation, and stochastic scenario-decomposition techniques (e.g., Progressive Hedging) have proven effective for specific classes of large-scale optimization problems, their application to the present model is challenged by its multi-objective structure, tightly coupled material flows, and possibilistic (non-scenario-based) uncertainty representation. Consequently, general-purpose multi-objective metaheuristics were chosen to provide a flexible and computationally tractable alternative for approximating the Pareto front in this context. Future research may explore hybrid solution strategies that integrate decomposition principles with metaheuristic frameworks to further enhance solution quality and scalability.

From a practical viewpoint, it offers a decision framework that translates technical configurations into actionable levers for growers, processors, and regulators. Limitations include the single-country case focus, static pricing of water and packaging externalities, and simplified behavioral responses. An additional limitation of the presented approach is the Normal approximation adopted for the olive quality parameters. As biological yields can be negatively skewed through production function concavity (see Hennessy, 2009), a natural extension of this research is skew-aware modeling that would replace the Normal assumption with skewed or distribution-free representations of yield quality. Future research may also extend the proposed strategic model to a multi-period dynamic framework that aligns with perishability, time-dependent quality degradation, as well as inventory decay through time-dependent variables and parameters, building on rolling-horizon rescheduling approaches developed for agricultural harvest (Montenegro-Dos Santos et al., 2023). This would provide deeper insight into operational decisions beyond the structural network configuration analyzed in this study. Additionally, incorporating endogenous pricing, regulatory instruments, and cross-commodity benchmarking could help generalize and strengthen the stability-first playbook identified here.

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